Fair dispatching with price-responsive couriers: an end-of-day earnings guarantee for the meal delivery routing problem

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Abstract

We study dispatching problems in which a platform repeatedly offers work and pay to self-employed agents who may reject it, and is accountable for how their earnings are distributed at the end of the day rather than within each decision window. In meal delivery, a platform offers routes and pay to crowdsourced couriers every two minutes, couriers accept according to the pay per kilometre, and fairness-aware dispatching has so far balanced work within the window under perfect acceptance. We formulate the problem as a constrained Markov decision process whose state carries an equity history per courier, whose constraint bounds the end-of-day Gini coefficient of hourly earnings over the couriers connected for at least one hour, idle ones included, and whose transition draws acceptance from a logit on pay per kilometre under two observed regimes, manual and auto-accept. We develop three cost function approximations that read the history every round, including a per-window mixed-integer programme with a linearisation of the Gini constraint that is exact by blocks of courier pairs, and compare them with four benchmarks on 24 operating days from six Colombian cities, with acceptance estimated from field interviews and a survey. The central finding is a question of population: the courier-centric dispatcher ends the day at a Gini coefficient of 0.30 by giving work to 97% of the eligible couriers, whereas every order-centric policy ends between 0.52 and 0.57 under manual acceptance because it leaves a quarter of them idle, while the usual statistic over active couriers tells the opposite story. Reading exposure over the eligible couriers and offering the most-deprived couriers to the solver closes about half of that gap at no coverage cost.

Keywords: Routing, Meal delivery, Fairness, Constrained Markov decision process, Crowdsourced couriers

1. Introduction

Dispatching problems arise when a limited set of agents must be matched repeatedly to requests that appear over time. In a corporate fleet, the agents are vehicles: an assignment is executed as

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soon as it is made, and the quality of a dispatching policy is measured by the service it delivers to customers (Dantzig and Ramser 1959, Pillac et al. 2013). In platform operations, however, the agents are self-employed workers who decide whether to execute each assignment and who earn their living from the sum of the assignments they execute over a day. Two features of such settings are absent from the corporate model. First, an assignment is an offer: whether it is accepted depends on what it pays relative to the effort it asks for, so the transition of the system is endogenous to the pay decision. Second, the platform is accountable for a distributional outcome, the earnings of its workers at the end of the day, which no single decision window determines but every decision window affects.

Meal delivery is the setting in which these features are sharpest. On-demand platforms coordinate restaurants, customers and a crowd of couriers through an assignment decision repeated every few minutes. The operational literature on the meal delivery routing problem (MDRP) has made that decision efficient: rolling-horizon re-optimisation, order bundling and anticipatory assignment deliver high coverage with short click-to-door (placement to delivery) times (Reyes et al. 2018, Yıldız and Savelsbergh 2019, Ulmer et al. 2021). The couriers' side has received less attention: a courier earns over a day, not over a two-minute window, so a dispatcher that balances work within each window can still end the day with very unequal hourly earnings when couriers connect at different times or some systematically receive the better routes. Moreover, a courier is not a vehicle: in the field, couriers reason in pay per kilometre and reject long, low-paying routes, and, in the platform we study, the workforce has moved to an auto-accept regime, backed by sanctions, because manual rejection erodes coverage (Section 3.1). Similar structures arise in ride-hailing (Lesmana et al. 2019), in crowdsourced last-mile delivery (Sobhanan et al. 2024) and in gig platforms that guarantee an hourly wage (Nair et al. 2022): a platform decides who receives which work and at what pay, workers respond to that pay, and the fairness outcome is defined over a horizon longer than the decision window.

Methodologically, these problems are challenging because the fairness guarantee, the acceptance behaviour and the routing decision are coupled within the same sequential process: the value of offering a route depends on whether it will be accepted, which depends on the pay, and the earnings accumulated by every courier determine whether the end-of-day guarantee can still be met. Anticipation is therefore required, not of future demand, which rolling-horizon dispatching handles procedurally, but of the distributional consequence of the current assignment at the end of the day: who receives each route and what the offer pays must be decided window by window while keeping track of what every courier has earned and worked so far.

We address this problem as a constrained Markov decision process (CMDP, Altman 1999). The state carries, in addition to the positions of couriers and the pool of orders, an *equity history* $\mathcal{H}_t$ with the earnings, orders and busy time accumulated by every courier. The constraint bounds by a tolerated inequality α the end-of-day Gini coefficient of hourly earnings over the *eligible* couriers, those connected for at least one hour, with the idle ones counted at zero. The transition draws

the acceptance of every offer from a logit on pay per kilometre, estimated from field data, with a manual regime in which couriers reject and an auto-accept regime in which they do not. We then develop three policies that read $\mathcal{H}_t$ when deciding the current window: **BRKGA-Fair** biases the decoder of a biased random-key genetic algorithm by each courier's equity deficit, **Pay-BRKGA** adds a pay gene per courier that sets the offer within a bounded band around the tariff, and **MIP-Fair** solves a per-window mixed-integer programme (MIP) in which the Gini constraint is written on the projected end-of-window history and linearised exactly by blocks of courier pairs. In terms of the framework of Powell (2021), all three are cost function approximations: they modify the problem solved in each window with terms computed from $\mathcal{H}_t$ instead of learning a value of the future.

Three research questions guide the study. First, does an end-of-day fairness guarantee require the equity history in the state, or can fairness enforced within each window deliver it? Second, which lever satisfies the guarantee at the lowest cost in coverage and punctuality, choosing who receives each route or choosing what the offer pays, and does the answer depend on the acceptance regime? Third, under which operational conditions, acceptance regime, logit threshold, tolerated inequality and fleet composition, is the guarantee cheap or expensive, and which α is sustainable for a platform? Using 24 operating days from six Colombian cities with acceptance calibrated in the field (135 surveys, 44 interviews), we compare the three policies against a corporate greedy dispatcher, a coverage integer linear programme (ILP), a coverage-only BRKGA and a hierarchical policy that is fair within each window. The computational study establishes the following insights:

- The guarantee is a property of the day, not of the window: a policy that is fair within every window (**Hierarchical**) ends the day more unequal than **Greedy** (0.53 against 0.30); among the order-centric policies, only those that read exposure and offer the most-deprived couriers to the solver halve that gap.
- Coverage is identical across policies under auto-accept and within 1.2 points of **Greedy** under manual acceptance for every policy but the coverage ILP, so the cost of the guarantee is paid in *who* receives each route and in lateness (0.5–1.8 minutes).
- The acceptance regime matters more than the policy for coverage: the regime loss is 1.6–4.8 points against 1.2 between **Greedy** and the best order-centric policy (3.2 including the coverage ILP).
- The pay lever cuts the bill rather than inequality: **Pay-BRKGA** pays 12% (manual) to 20% (auto-accept) less than the tariff at the same coverage under auto-accept and 1.7 points less under manual acceptance, a transfer from couriers who already earn below the minimum wage to the platform.
- Dispatching alone cannot lift couriers to the statutory minimum wage at the current tariff (97–100% remain below it under every policy, 92% in the survey); what it can do is distribute the shortfall evenly.
- The three criteria for α give the interval $[0.30, 0.45]$ rather than a point: 0.30 is the status

quo of **Greedy**, which no order-centric policy reaches, and 0.45 the operating point of the order-centric policies, above the inequality band; choosing between them is a managerial trade-off of a point of coverage and one to two minutes of lateness against ten points of Gini.

The paper makes three contributions. First, as a modelling contribution, we formulate meal-delivery dispatching as a CMDP with an intertemporal equity history in the state, an end-of-day Gini constraint and price-endogenous acceptance. The constraint slot admits the corporate model ($\alpha = 1$), workload-balance criteria and minimum-wage criteria, and a numerical example illustrates why per-window fairness does not imply end-of-day fairness. Second, as a methodological contribution, we develop three cost function approximations for the same CMDP, including a linearisation of the Gini constraint on projected values that is exact by blocks of courier pairs and keeps the per-window programme within a 10-second budget, and we verify them against exhaustive enumeration on a toy instance. Third, as a managerial contribution, we measure the cost of the fairness guarantee on 24 real operating days from six cities with field-calibrated behaviour: the frontiers per policy and regime, the loss attributable to the acceptance regime, the sensitivity to the behavioural threshold, the choice of α and the side effect on emissions. To the best of our knowledge, this is the first MDRP formulation that combines an end-of-day inequality constraint over a population the dispatcher cannot shrink, an equity history in the state and courier acceptance that responds to the pay the platform decides.

The remainder of this paper is organised as follows. Section 2 reviews the related literature and Section 3 defines the problem. Section 4 presents the three policies, Section 5 the experimental setting and Section 6 the computational study. Section 7 synthesises the findings and Section 8 concludes; the Appendix provides additional detail on the literature, the formulation, the algorithms and the results.

2. Related work

The problem studied in this paper lies at the intersection of three literature streams: the operational stream on the MDRP (routing and assignment), the stream on fairness and compensation in delivery and ride-hailing platforms, and the behavioural stream on courier acceptance and availability. We review the three streams, which motivate our benchmark policies, and then the methodological positioning of the proposed policies; Table 1 summarises the positioning and Appendix A provides a broader review.

Problem-related literature. The MDRP was defined by Reyes et al. (2018) as a rolling-horizon, dynamic-deterministic problem: at each re-optimisation epoch a deterministic instance is solved and the horizon rolls forward as new orders arrive, so uncertainty is handled procedurally rather than through anticipation. Their public instance library (Grubhub) has become the standard benchmark. Yıldız and Savelsbergh (2019) obtained provably high-quality solutions for a deterministic variant. Steever et al. (2019) revise assignments whenever an order arrives or traffic changes;

	Paper	Courier welfare		Courier behaviour and pay			Real data
		Fairness metric	Guarantee horizon	Acceptance	Pay decided	History in state	
MDRP operations	Reyes et al. (2018)			perfect			✓
	Yıldız and Savelsbergh (2019)			perfect			✓
	Steever et al. (2019)			perfect			
	Ulmer et al. (2021)			perfect			✓
	Liu and Luo (2023)			perfect			✓
	Auad et al. (2023)	satisfaction	shift	perfect			✓
	Simoni and Winkenbach (2023)			perfect			
	Neria et al. (2025)			perfect			
Fairness and pay	Wen et al. (2010)	workload	horizon	perfect			
	Liao et al. (2020)	workload	instance	perfect			
	Lesmana et al. (2019)	income Gini	window	perfect			✓
	Gupta et al. (2022)	hourly-income gap	day	perfect			✓
	Nair et al. (2022)	hourly wage	shift	perfect	idle pay		✓
	Martínez-Sykora et al. (2024)	orders range	shift	perfect			✓
	Sobhanan et al. (2024)	workload	instance	perfect			
Behaviour	Ahamed et al. (2021)			exogenous			
	Bozanta et al. (2022)			exogenous			
	Quintero Rojas (2020)			distance			✓
	TODO authors (2025)	[TODO:]	reward term	[TODO:]	[TODO:]	[TODO:]	[TODO:]
	Our work	hourly-earnings Gini	day	pay per km	✓	✓	✓

Table 1: Positioning of our work relative to the literature across courier welfare, courier behaviour and pay, and data. Acceptance: "perfect" when every assignment is executed, "exogenous" when availability is random but independent of the offer, otherwise the driver of the acceptance model; guarantee horizon: the span over which the fairness measure is evaluated. [TODO: complete the row of TODO authors (2025) (placeholder bib entry).]

Ulmer et al. (2021) delay assignment until preparation-time uncertainty resolves and anticipate it with approximate dynamic programming; Liu and Luo (2023) dispatch and route on-demand deliveries from stores with random demand. Later contributions address shared logistics (Bi et al. 2023), courier satisfaction through dynamic operating regions (Auad et al. 2023), order batching for crowdsourced fleets (Simoni and Winkenbach 2023) and ghost kitchens (Neria et al. 2025). For the broader class of stochastic dynamic vehicle routing problems we refer to Soeffker et al. (2022) and Hildebrandt et al. (2023), and for their modelling as sequential decision processes to Ulmer et al. (2020). Across these models, acceptance is perfect or exogenous and couriers are treated as vehicles. This literature motivates the **Greedy** and **ILP** benchmarks of Section 5.2: the platform's practice as the couriers describe it, and the exact per-window coverage dispatcher that **MIP-Fair** extends.

Fairness has entered delivery and ride-hailing platforms mostly as *balance*. Wen et al. (2010) balance workload and income opportunities within a multi-period routing model, and Liao et al. (2020) balance workload in meal delivery with a two-stage evolutionary algorithm (NSGA-II), arguing that an equitable distribution is itself an incentive. Lesmana et al. (2019) define fairness as income inequality among ride-sourcing drivers and trace the efficiency–fairness frontier with a reassignment algorithm applied batch by batch. FairFoody (Gupta et al. 2022) and WORK4FOOD (Nair et al. 2022) treat fairness as income distribution among couriers in Indian food-delivery data: the former balances earnings by assignment, the latter guarantees a minimum hourly wage by controlling the number of active couriers and paying for idle time. Martínez-Sykora et al. (2024)

explore fairness constraints in food-delivery routing and scheduling on UK data and find moderate efficiency costs, and Sobhanan et al. (2024) give bi-objective equity–cost frontiers for crowdsourced last-mile delivery. These works share two limits that this paper removes: fairness is enforced per window or per instance, or amortised over the day through an objective or a reward term rather than constrained at its end, and couriers accept whatever is assigned. The per-window logic motivates our **Hierarchical** benchmark, which balances the pay offered among the free couriers of each window and ignores the history (Section 5.2). On the economic side, Reyes et al. (2018) and Yıldız and Savelsbergh (2019) include a shift-level guaranteed minimum set by the company, Bajec and Tuljak-Suban (2022) use Shapley-value incentives for sustainable crowdshipping, and the algorithmic-fairness literature shows that short- and long-horizon fairness can diverge (Corbett-Davies et al. 2023). None of these integrates pay as a dispatching decision to which acceptance responds. The Gini constraint itself belongs to inequity-averse optimisation (Karsu and Morton 2015, Ogryczak 2000); in ride-hailing, Sühr et al. (2019) and Xu and Xu (2020) balance driver income over repeated matchings, again under perfect acceptance.

Courier behaviour has been modelled mainly as availability. Reinforcement-learning approaches (Ahamed et al. 2021, Bozanta et al. 2022) include courier availability in the state but keep acceptance exogenous, and Haferkamp et al. (2024) address repositioning. The closest precedent to endogenous acceptance is the discrete-event framework of Quintero Rojas (2020), in which the probability of accepting a notification depends on the assignment distance and a courier-level parameter, but not on the pay offered. Customer acceptance has been anticipated in delivery routing (Ulmer and Thomas 2020), but there it is the platform that accepts requests, not the agent who accepts work. No MDRP study models acceptance as a function of the offered pay, although a courier offered a long, low-paying route is more likely to reject it, which biases both coverage and fairness estimates. The empirical literature on gig work documents the two levers we model: pay shapes the supply of self-scheduling workers (Cachon et al. 2017, Allon et al. 2023), and crowdsourced delivery has been planned with occasional drivers whose participation and compensation are decisions (Archetti et al. 2016, Dayarian and Savelsbergh 2020, Ulmer and Savelsbergh 2020). [TODO: One sentence on what TODO authors (2025) (arXiv:2501.05808) actually does, after reading it.]

Methodological literature. Our policies belong to the class of cost function approximations in the framework of Powell (2021) and Powell (2019): the problem solved in each window is modified by terms, the equity deficit, the projected Gini excess and the pay excess, that encode the end-of-day consequence of the current decision, without learning a value function. Constrained MDPs are usually solved with Lagrangian or primal–dual methods (Altman 1999, Borkar 2005); our constraint is on a terminal non-linear functional, outside that machinery (Section 3.2), which is why we use cost function approximations. This is the class of parametrised policies of the dynamic routing literature (Soeffker et al. 2022), whose parameters are tuned by enumeration or learned by simulation (Stein et al. 2025); we tune ours over a grid and report the full frontier, since the trade-

off itself is the object of study. Two methodological components are specific to this paper. The first is the biased random-key genetic algorithm (Gonçalves and Resende 2011) with rolling horizon that we ported from our earlier work on the MDRP (Giraldo-Herrera and TODO co-authors 2026, Giraldo-Herrera and Álvarez 2024); the decoder is the component that reads $\mathcal{H}_t$. The second is the linearisation of the Gini constraint, which in its natural form requires one variable per pair of couriers; because a courier takes at most one route per window, only the pairs of free couriers need variables, and we verify the reformulation against the all-pairs model. The Gini coefficient is used in its mean-absolute-difference form (Yitzhaki 1979). To the best of our knowledge, the proposed policies are the first to be evaluated against an end-of-day inequality constraint on real meal-delivery operations with acceptance that responds to the pay the platform decides.

3. Problem definition

In this section, we define the fair meal-delivery dispatching problem and formulate it as a constrained sequential decision process. We first describe the problem and the field evidence on courier behaviour that shapes it. We then define the components of the process. An illustrative numerical example connecting the formulation to the dispatching decisions, and showing why per-window fairness does not imply end-of-day fairness, is provided in Appendix B.2.

3.1. Problem description

A platform connects a set of stores R, a set of users U and a set of couriers C over one operating day. Stores are known in advance; couriers are revealed as they connect (position and vehicle at log-in) and disconnect; orders are revealed at placement time. Every two minutes, the dispatcher assigns to each available courier at most one route, where a route collects up to Q orders from a single store and delivers them to their users, and offers it with a pay. The courier accepts or rejects the offer. Accepted routes are executed (pick-up, drop-off); the orders of rejected routes return to the pool for the next window. Couriers are notified only while idle or picking up. An order is cancelled, and counted against coverage, when no feasible assignment exists before its hard service deadline. We assume that each user places at most one order per day and that a cancelled order does not re-enter the system, as in the public MDRP instances (Reyes et al. 2018).

Three aspects distinguish this problem from the corporate MDRP, and each is supported by field evidence collected with couriers of the platform in Bogotá (44 semi-structured interviews in 2020–2021 and an intercept survey of 135 couriers in May–June 2021; the full assumption-versus-evidence table is in the online supplement). First, *acceptance responds to pay per kilometre.* Couriers reason in pay per kilometre ("nobody takes 4.1 km for 3,300"); in the survey, pay (importance 5.71 on a 7-point scale) and distance (5.65) are the dominant factors in that order. A logit on ln(pay/km) fitted to 32 acceptance decisions by 11 interviewees gives an indifference threshold of 903 Colombian pesos (COP) per kilometre (95% interval [615; 1,242]) and slope 2.89 ([2.02; 5.23]). Second, *there are*

two decision regimes. In the 2020 interviews no courier auto-accepted and rejection was common; by 2021 most interviewees and 64% of surveyed couriers auto-accepted, behind which are platform sanctions for rejecting (60% of surveyed couriers had been blocked). In the auto-accept regime the courier's decision is whether to be connected at all, which the data record as on and off times. Third, *the tariff is flat in distance.* The per-kilometre rate estimated from the interviews is indistinguishable from zero (155 COP/km, 95% interval $[-103; 413]$); the survey reports 2,200 COP per order outside the platform's bonus ladder. We therefore model a flat tariff and treat per-offer pay as a policy instrument, not as a description of current practice.

The evidence also fixes the fairness outcome. Median hourly earnings while connected were 2,622 COP in the survey and 3,957 COP in the interviews; 91.9% of surveyed couriers earned less than the 2021 statutory minimum wage per hour (4,368 COP), and the platform guarantees no hourly pay. The Gini coefficient of daily income among surveyed couriers is 0.39 outside the bonus ladder and 0.26 inside it. We therefore measure fairness on hourly earnings at the end of the day, normalised by connected time so that connecting late is not penalised, over the couriers connected for at least one hour, with the idle ones counted at zero so that the dispatcher cannot improve the statistic by leaving couriers without work, and we use the two survey values as anchors for the tolerated inequality. Service and decision times are taken from the same sources: a median in-store wait of 20 minutes with a purchased order (60 minutes in the stressed interview accounts) and a median decision time of 16 seconds for an offer displayed for 20 seconds.

The goal of the platform is to find a dispatching and pay-setting policy that minimises expected undelivered orders and lateness over the day while guaranteeing that the end-of-day Gini coefficient of the eligible couriers' hourly earnings does not exceed a tolerated value α.

3.2. Sequential decision process

The problem is stochastic and dynamic: orders are revealed over the day, preparation times are uncertain, and the acceptance of every offer is a random variable whose distribution depends on the pay the platform decides. Because assignment and pay decisions are made every two minutes and change the earnings and availability of couriers for the rest of the day, we model the problem as a sequential decision process (Powell 2021, Ulmer et al. 2020, Cuellar-Usaquén et al. 2024) with a constraint on a terminal quantity, a constrained Markov decision process (Altman 1999). In the following, we define its components: decision point and state, decisions, exogenous information and transition, cost function, and the constrained objective.

Decision point and state. The process consists of decision points $t \in T$ with $T = \{0, 1, \ldots, t_{\max}\}$, one at the start of every two-minute window of the day; we refer to the dispatching decision made at t as a *round.* At decision point t, the state S_t contains the information accessible to the dispatcher: the couriers, the orders waiting to be assigned, their preparation, and what every courier has earned and worked so far. Its components are:

- the courier information $\mathcal{L}_t$, giving for each courier $c \in C$ its position $\ell_c(t)$, status $\sigma_c(t) \in \{\text{idle, picking-up, in-transit, dropping-off, offline}\}$ and time to next availability $\eta_c(t)$,
- the set $\mathcal{O}_t$ of unassigned orders; order o is linked to a store $r_o \in R$ with location ℓ_{r_o} and to a user $u_o \in U$ with drop-off location ℓ_{u_o}, and has placement time a_o, ready time e_o and promised drop-off time d_o,
- the vector $\mathcal{P}_t$ of estimated remaining preparation times $\hat{p}_o(t)$ of the unassigned orders,
- and the equity history $\mathcal{H}_t = (h_c(t))_{c \in C}$, with

$$h_c(t) = \big(\text{earn}_c(t),\ \text{ord}_c(t),\ \text{busy}_c(t)\big), \qquad c \in C, \tag{1}$$

the earnings, the orders delivered and the busy time (minutes in motion or serving) accumulated by courier c up to t; the connected time $\text{conn}_c(t) = t - e_c$ follows from the on-time recorded in $\mathcal{L}_t$, and the hourly earnings so far are $w_c(t) = \text{earn}_c(t)/\text{conn}_c(t)$. In summary, $S_t = (\mathcal{L}_t, \mathcal{O}_t, \mathcal{P}_t, \mathcal{H}_t)$, with $\mathcal{O}_0 = \emptyset$ and $h_c(0) = (0, 0, 0)$. The equity history is what enables dynamic fairness control, since a policy that reads $\mathcal{H}_t$ can prevent inequities from compounding across windows; it is absent from existing MDRP formulations, where the state contains the positions and plans of couriers but not what they have earned (Ulmer et al. 2020).

Courier c has on-time e_c, off-time $l_c > e_c$, initial location ℓ_c and vehicle type $v_c \in V$, $V = \{\text{walking, bicycle, motorcycle, car}\}$. Travel times are deterministic, $\tau(\ell, \ell') = \kappa \, \|\ell - \ell'\| / \nu(v_c)$, with a circuity factor κ and a vehicle-dependent speed $\nu(v_c)$; service times s_r at the store and s_u at the user are included in the duration of a route. A route s collects at most Q orders from a single store and is served by a single courier; $\text{dist}(s)$ is its length, τ_s its duration and $|s|$ the number of orders it contains.

Decisions. At each decision point t, the dispatcher decides which available courier receives which route and what the offer pays, a set of triples

$$x_t \subseteq \big\{(c, s, \rho_{c,s}) : c \in C_t^{\text{free}},\ s \in \mathcal{S}_t\big\}, \tag{2}$$

where C_t^{free} is the set of available couriers (idle, or picking up and able to receive a further offer), $\mathcal{S}_t$ the set of feasible routes that can be built from $\mathcal{O}_t$, and $\rho_{c,s}$ the pay offered for the pair, which is paid in full if the offer is accepted and not otherwise: $\text{pay}(c, s, \rho_{c,s}) = z_{c,s}\, \rho_{c,s}$, with $z_{c,s}$ the acceptance indicator defined below. A route is feasible for a courier if every order can be picked up after its ready time and dropped off before its hard deadline $d_o + \Delta^{\max}$, given the courier's position and speed. A decision $x_t \in \mathcal{X}(S_t)$ is feasible if each courier receives at most one route and each order appears in at most one route.

The pay offer follows the platform's tariff, an affine function of route distance,

$$\rho_{c,s} = \rho_0 \cdot |s| + \rho_1 \cdot \text{dist}(s), \tag{3}$$

with ρ_0 a base fee per order and ρ_1 a per-kilometre rate; following the field evidence of Section 3.1, the baseline tariff is flat ($\rho_0 = 2{,}200$ COP, $\rho_1 = 0$). Pay-setting policies may scale the offer by a multiplier $m_c \in [1-\delta, 1+\delta]$ around the tariff, not set it arbitrarily.

Acceptance, exogenous information and transition. After decision x_t in state S_t, two kinds of information are revealed. The first is the acceptance of every offer: courier c accepts route s offered at pay $\rho_{c,s}$ with probability

$$p_{c,s}(\rho_{c,s}) = \begin{cases} \sigma\big(b\,[\ln(\rho_{c,s}/\operatorname{dist}(s)) - \ln\theta]\big) & \text{manual regime,} \\ 1 & \text{auto-accept regime,} \end{cases} \tag{4}$$

where σ is the logistic function, θ the pay per kilometre at which acceptance is even and b the slope on log pay per kilometre; this is the form the field evidence supports, in which the pay decided by the platform enters the decision directly, unlike the distance-only acceptance of Quintero Rojas (2020). We denote by $\mathbf{z}_t \in \{0,1\}^{|x_t|}$ the vector of Bernoulli acceptance realisations. The second kind of information is exogenous to the decision: $\omega_{t+1} = (\Delta\mathcal{O}_{t+1}, \Delta\mathcal{P}_{t+1}, \Delta\mathcal{A}_{t+1})$ reveals the orders placed during the window, the realised preparation times, and the courier log-ins and log-outs. The transition function $\mathcal{T}(S_t, x_t, \mathbf{z}_t, \omega_{t+1})$ leads to the new state S_{t+1}:

- accepted routes are executed: the courier's status, position and time to availability in $\mathcal{L}_{t+1}$ follow from the route's pick-up and drop-off sequence, and the orders of accepted routes leave $\mathcal{O}_{t+1}$,
- the orders of rejected routes remain in $\mathcal{O}_{t+1}$, together with the new orders $\Delta\mathcal{O}_{t+1}$; orders whose hard deadline can no longer be met are cancelled,
- preparation estimates are replaced by realisations where revealed, $\mathcal{P}_{t+1}$,
- couriers that log in or out change status according to $\Delta\mathcal{A}_{t+1}$,
- and the equity history is updated: the pay of an accepted route is credited at acceptance, $\text{earn}_c(t+1) = \text{earn}_c(t) + z_{c,s}\,\rho_{c,s}$, its orders are added to ord_c as they are delivered, and busy_c accumulates the minutes the courier spends in motion or serving.

Cost function. The cost $\mathcal{C}(S_t, x_t, \mathbf{z}_t, \omega_{t+1})$ of a decision penalises orders that are cancelled during the window and the lateness of orders delivered during the window,

$$\mathcal{C}(S_t, x_t, \mathbf{z}_t, \omega_{t+1}) = c_1 \sum_{o \in \mathcal{O}_t} \mathbb{I}[o \text{ cancelled}] + c_2 \sum_{o \text{ delivered}} \max\big(0,\ \tau_{\text{drop}}(o) - d_o\big), \tag{5}$$

where $\mathbb{I}(\cdot)$ is the indicator function and $c_1, c_2 > 0$ weigh coverage against punctuality. Its extension with a payment term, used by the pay-setting policy, is

$$\mathcal{C}^{+}(S_t, x_t, \mathbf{z}_t, \omega_{t+1}) = \mathcal{C}(S_t, x_t, \mathbf{z}_t, \omega_{t+1}) + \lambda \sum_{(c,s) \in x_t} z_{c,s}\ \text{pay}(c, s, \rho_{c,s}), \tag{6}$$

where $\mathrm{pay}(c, s, \rho_{c,s})$ is the amount paid for an accepted route and $\lambda > 0$ trades platform cost against coverage. In our experiments coverage and lateness are reported separately; the lexicographic reading coverage $\succ$ lateness used by all policies corresponds to $c_1 \gg c_2$. [TODO: state the values of c_1 and c_2 if a scalar cost is reported.]

Fairness metrics and constrained objective. Let $w_c = \mathrm{earn}_c(t_{\max})/\mathrm{conn}_c(t_{\max})$ be the hourly earnings of courier c at the end of the day, zero if it delivered nothing. The population over which fairness is measured is the set of *eligible* couriers $C^e = \{c \in C : \mathrm{conn}_c(t_{\max}) \geq 1 \text{ h}\}$, fixed by the couriers' connection decisions and not by the dispatcher. The primary fairness metric is the Gini coefficient of hourly earnings over C^e,

$$F_{\mathrm{Gini}}(\mathbf{w}) = \frac{\sum_{c \in C^e} \sum_{c' \in C^e} |w_c - w_{c'}|}{2\,|C^e| \sum_{c \in C^e} w_c}, \qquad \mathbf{w} = (w_c)_{c \in C^e}, \tag{7}$$

with $F_{\mathrm{Gini}} \in [0, 1]$ and 0 perfect equality. Normalising by connected time makes Eq. (7) an exposure-adjusted statistic, and the two restrictions on the population make it a guarantee: a policy cannot lower it by leaving couriers idle, since they enter with $w_c = 0$, nor by dispatching single orders to couriers who connect for a few minutes, since they are excluded. The same statistic over the couriers who delivered at least one order, $F^{\mathrm{act}}_{\mathrm{Gini}}$, is the one used by the fairness literature of Section 2; it can be lowered by both manoeuvres, and Section 6.1 shows that it is. Three secondary metrics are reported over the active couriers: max–min fairness $F_{\mathrm{MaxMin}} = \min_c w_c$, order-count balance F_{Orders}, the Gini coefficient of orders delivered per connected hour, and the minimum-wage deficit F_{Deficit}, the share of active couriers below the statutory minimum hourly wage w^*.

A solution is a policy $\pi \in \Pi$ that assigns a decision $\mathcal{X}^\pi_t(S_t)$ to each state; the optimal policy minimises the expected cumulative cost subject to the end-of-day guarantee:

$$\pi^* = \operatorname*{argmin}_{\pi \in \Pi} \mathbb{E}\left[\sum_{t \in T} \mathcal{C}\big(S_t, \mathcal{X}^\pi_t(S_t), \mathbf{z}_t, \omega_{t+1}\big) \,\Big|\, S_0\right] \tag{8}$$

$$\text{s.t.} \quad \mathbb{E}\big[F_{\mathrm{Gini}}(\mathbf{w}^\pi) \,\big|\, S_0\big] \leq \alpha, \tag{9}$$

$$\mathcal{X}^\pi_t(S_t) \in \mathcal{X}(S_t) \qquad \forall t \in T, \tag{10}$$

where $\alpha \in [0, 1]$ is the tolerated inequality and $\mathbf{w}^\pi$ the end-of-day hourly earnings induced by π; the day is finite, so no discounting is applied. Eq. (9) is deliberately written on an end-of-day quantity, not per decision point, which is what the equity history makes controllable; in the computational study we report, besides the mean of F_{Gini}, the fraction of realisations in which $F_{\mathrm{Gini}} \leq \alpha$ holds, which is the chance-constrained reading $\mathbb{P}(F_{\mathrm{Gini}} \leq \alpha) \geq 1 - \beta$ of Eq. (9); we report it for every α of the grid rather than fixing β. Eq. (9) constrains a terminal, non-linear functional of the whole trajectory rather than an expected cumulative cost, so the Lagrangian machinery of constrained MDPs (Altman 1999, Borkar 2005) does not apply directly; the policies of Section 4 are heuristics

for either reading. A minimum-wage constraint $\mathbb{E}[w_c^\pi] \geq w^*$ is not imposed because 92% of couriers earn below w^* in the field data, so it would be infeasible on every instance; it is pursued through F_{Deficit} by the pay-setting policy (Section 4.2) and reported for every policy.

The formulation relates to three familiar models: with $\alpha = 1$, perfect acceptance and tariff pay, Eqs. (8) - (10) are the corporate, efficiency-only MDRP of Reyes et al. (2018); the constraint slot of Eq. (9) admits F_{Orders} in place of F_{Gini}, the balance criterion of Wen et al. (2010) and Liao et al. (2020); and a bound on F_{Deficit}, or the minimum-wage constraint alone, is the wage criterion of Nair et al. (2022), whose levers (fleet size and paid idle time) this model does not have. Since the vehicle type of every courier is recorded, the emissions of a decision, $E(x_t) = \sum_{(c,s)\in x_t} z_{c,s}\, \epsilon(v_c)\, \text{dist}(s)$ with $\epsilon(v)$ a per-kilometre emission factor (zero for walking and bicycles), are a diagnostic reported alongside the frontiers, not a constraint.

Even in the two-courier example of Appendix B.2 the challenge becomes evident: a policy that equalises the pay offered within every window can end the day with a Gini coefficient of 0.5 between two equally connected couriers where 0 was attainable at the same coverage, because the per-window statistic ignores what each courier has already earned. A policy therefore needs $\mathcal{H}_t$ to control Eq. (9); the next section presents three policies that read it.

4. Method

In this section, we present our policies. We first give a motivation and overview of the shared architecture and then present the individual policies in detail.

4.1. Motivation and overview

Effective decision making is challenging for two reasons.

- **Search:** A round decides, for up to a few hundred free couriers in a zone (two thousand on the largest Bogotá day) and a few hundred ready orders, which courier receives which candidate route and what the offer pays, within the two-minute window.
- **Evaluate:** The value of a decision has two components that are revealed at different times. Whether a route is accepted is revealed within seconds and depends on the pay; whether the end-of-day guarantee is met is revealed only at the end of the day and depends on every subsequent round. The second component cannot be evaluated per window: as the example of Appendix B.2 shows, two decisions that are indistinguishable by any statistic of the current window can lead to end-of-day Gini coefficients of 0 and 0.5.

We address these challenges as follows. All policies, proposed and benchmark, are implemented as the assignment step of one discrete-event simulator and share (i) a round every two minutes; (ii) the same candidate routes, every singleton and one bundle per store with its Q most urgent ready orders, a generator under which bundling is nearly switched off (mean bundle size 1.00–1.03, Section 6.3), and candidate pairs built in the same way: for each route, the k available couriers

nearest to its store in straight-line distance that can feasibly serve it ($k = 25$ for every policy except **MIP-Fair**, which uses $k = 8$ for budget reasons, Section 5.5) plus, when the deficit-aware prefilter is on, the k_d available couriers with the largest positive deficit outside that set, the nearest among ties, again subject to feasibility; (iii) the acceptance model of Eq. (4) and the flat tariff; and (iv) the same metrics, computed on the end-of-day history $\mathcal{H}_{t_{\max}}$. Let K_t denote the set of candidate (courier, route) pairs at round t. Each policy returns a subset of K_t with at most one route per courier and one route per order, together with a pay offer per pair, that is, a feasible decision x_t in the sense of Eq. (2).

The three policies differ in how they use $\mathcal{H}_t$ to steer the current round towards the end-of-day guarantee, and in which lever they move. **BRKGA-Fair** keeps the population and fitness of a coverage-maximising genetic algorithm and biases its decoder: among decodings of equal coverage, routes are redirected to couriers whose hourly earnings so far fall below the mean, by a weight γ. **MIP-Fair** reads the constraint literally: in every round it solves a mixed-integer programme that maximises coverage first and then minimises the amount by which the projected end-of-round Gini coefficient exceeds α, with the constraint linearised exactly. **Pay-BRKGA** moves the other lever: it keeps the decoder of **BRKGA-Fair** without the deficit bias and adds one pay gene per courier, so that the offer can rise for couriers below the minimum wage and fall where acceptance is not at stake, within a band around the tariff; its fitness trades expected coverage, projected minimum-wage deficit and pay excess with a weight λ_{pay}. In terms of the framework of Powell (2021), the three are cost function approximations: the per-round problem is modified by terms computed from the state, the deficit, the projected Gini excess and the pay excess, rather than by a learned value of the future. The parameters γ, α and λ_{pay} are tuned by enumeration, since their frontier is what the platform needs to see. Algorithm 1 summarises a round for the three policies; the following sections detail each component.

Content and population of the history. Each policy reads $\mathcal{H}_t$ through a courier value and a population, and we evaluate two readings. The *activity basis* (the first generation of Section 6.1) values a courier by its earnings per busy hour, $\text{earn}_c(t)/\,\text{busy}_c(t)$, over the couriers of the round; the *exposure basis* (second generation) uses the earnings per connected hour $w_c(t) = \text{earn}_c(t)/\text{conn}_c(t)$ of Eq. (7), over the couriers connected for at least one hour so far, the eligible population as known at t, and switches on the deficit-aware prefilter with $k_d \in \{8, 25, 50\}$. We call the first-generation configuration from which a second-generation variant descends its *parent*. The deficit of a courier that has earned nothing is one under either basis, however long it has waited, so the prefilter puts the nearest idle couriers in front of the solver; the basis matters for the couriers with earnings, whom the activity basis rates by productivity and the exposure basis by what they have earned over their connected time.

Algorithm 1 Dispatching round shared by the three proposed policies

for $t = 0, \ldots, t_{\max}$ **do**
  build the candidate routes $\mathcal{S}_t$ (singletons and one bundle per store) from the ready orders in $\mathcal{O}_t$
  build the candidate pairs K_t: for each route, the k nearest available couriers that can feasibly serve it, plus the k_d available couriers with the largest positive deficit outside that set ($k_d = 0$ in the first generation)
  compute from $\mathcal{H}_t$ (Eq. (1)) the hourly earnings so far $w_c(t)$ of every connected courier
  if BRKGA-Fair or **Pay-BRKGA then**
    run the BRKGA for G generations with decoder of Algorithm 2 and score of Eq. (11) (fitness of Eq. (15) for **Pay-BRKGA**)
  else if MIP-Fair then
    solve the three-stage programme of Eqs. (12) - (14) over K_t
  offer the selected pairs with pay $\rho_{c,s}$; draw acceptance from Eq. (4)
  execute accepted routes, return rejected orders to the pool, update $\mathcal{H}_{t+1}$

4.2. Detailed description of the policies

In this section, we provide details of every component. We first describe the coverage-maximising BRKGA that **BRKGA-Fair** and **Pay-BRKGA** extend, then the three policies, and finally their verification.

Coverage BRKGA and decoder. The biased random-key genetic algorithm (Gonçalves and Resende 2011) with rolling horizon is ported from our earlier work on the MDRP (Giraldo-Herrera and TODO co-authors 2026). The chromosome has one key per ready order and one per available courier. The decoder (Algorithm 2 in Appendix B.3) visits stores in the order of their highest-key order; at each store it takes the head order, evaluates its candidate routes (singleton and bundle) against the K highest-key couriers among those that can feasibly serve the route, and keeps the pair with the best score. Fitness is coverage, with mean travel time as a tie-breaker. The decoder score is the finishing time per order, $\text{score}_{\text{fpo}}(s, c) = \text{score}_{\text{finish}}(s, c)/|s|$ with $\text{score}_{\text{finish}}$ the time at which the courier finishes the route, so that bundles compete with singletons; the original finishing-time score, under which a bundle never beats the singleton with the same head order, is an ablation in Appendix C.6. The decoder selects the K couriers among the candidate pairs of the route.

BRKGA-Fair. **BRKGA-Fair** is the coverage BRKGA with a decoder that reads the equity deficit from $\mathcal{H}_t$. Only the score changes:

$$\text{score}_{\text{fair}}(s, c) = \text{score}(s, c) - w \cdot \text{deficit}_c, \qquad \text{deficit}_c = \frac{\max(0,\ \bar{w}_t - w_c(t))}{\bar{w}_t} \in [0, 1], \tag{11}$$

where $w_c(t)$ is the courier's value under the basis in use (zero if it has earned nothing), $\bar{w}_t$ is the mean over the free couriers of the round, and γ is a weight in minutes per order: a courier with full deficit is treated as if every route of theirs finished γ minutes per order earlier under

$\text{score}_{\text{fpo}}$. The fitness that ranks the population is unchanged, so the deficit redirects routes among decodings of equal coverage and the population retains its maximum-coverage members; the deficit acts only among the K couriers that the decoder evaluates for each route, which bounds its effect by construction. With $\gamma = 0$ the policy is identical to the coverage BRKGA for the same seed, which isolates the value of the deficit term. A total-earnings basis would penalise connecting late rather than being under-served, and is reported as an ablation.

MIP-Fair. **MIP-Fair** solves, at each round, a mixed-integer programme over K_t with the Gini constraint of Eq. (9) written on the projected end-of-round values of $\mathcal{H}_t$. Let $x_{cs} \in \{0,1\}$ indicate that pair $(c,s) \in K_t$ is dispatched and $y_o \in \{0,1\}$ that order o is covered. Because a courier takes at most one route per round, the projected hourly earnings of a free courier are linear in x:

$$w_c = w_c^0\Big(1 - \sum_s x_{cs}\Big) + \sum_s x_{cs}\, w_c(s), \qquad w_c(s) = \frac{\text{earn}_c(t) + \rho_{c,s}}{(\tau_c^{\text{base}}(t) + \tau_s)/60}, \tag{12}$$

where τ_s is the route's travel-plus-service time in minutes, $\tau_c^{\text{base}}(t)$ is $\text{busy}_c(t)$ under the activity basis and $\text{conn}_c(t)$ under the exposure basis, and the projection assumes that the offer is accepted. Couriers that are busy, or connected for less than a short floor, enter with fixed values $w_c = w_c^0$. The Gini constraint is evaluated over all connected couriers, not only the free ones of the round, because Eq. (9) is over C^e and the free couriers are a biased subset; the free-only scope is reported as an ablation and turns out to be tighter but eight times more expensive (Appendix C.6). The two readings of Section 4.1 apply here as to the deficit of **BRKGA-Fair**; **Pay-BRKGA** reads the deficit only through the prefilter. Writing $n = |C_t|$ for the number of connected couriers, the constraint $F_{\text{Gini}}(\mathbf{w}) \le \alpha$ is equivalent to $\sum_{i<j} |w_i - w_j| - \alpha\, n \sum_c w_c \le 0$, and since the inequality is in the direction that allows it, it is linearised exactly as

$$\sum_{i<j} d_{ij} - \alpha\, n \sum_c w_c \;\le\; n\, W_{\text{ref}}\, v, \qquad d_{ij} \ge w_i - w_j,\;\; d_{ij} \ge w_j - w_i,\;\; v \ge 0, \tag{13}$$

where v is an elastic "Gini excess", the excess of the sum of absolute differences over $\alpha\, n \sum_c w_c$ scaled by nW_{ref} so that no round is infeasible, and $v^* > 0$ measures by how much α could not be met in that round. Building d_{ij} for all pairs is $O(n^2)$ per round, which is prohibitive with thousands of connected couriers: with 188 connected couriers the all-pairs model did not complete a two-hour slice of a Pereira day within four minutes of wall-clock time, and Bogotá would require 3.5 million pairs per round. The pairs are therefore handled in three blocks. Fixed–fixed pairs contribute a constant. Free–fixed pairs are linear in x, because the free courier's w_i takes one of finitely many values (w_i^0 or $w_i(s)$), so $|w_i - w_j| = (1 - \sum_s x_{is})\,|w_i^0 - w_j| + \sum_s x_{is}\,|w_i(s) - w_j|$. Only free–free pairs require the variables d_{ij}. The linearisation is exact for the projected values, which we verify against the all-pairs model on a toy instance; what is approximate is the projection itself: it assumes acceptance, it is taken over the couriers connected at t rather than over the end-

of-day population, the denominator of a dispatched courier advances by τ_s while the others' do not, and $\alpha \to 0$ minimises the sum of absolute differences of the projected values rather than their Gini coefficient, since the denominator $\sum_c w_c$ varies with x. The free–free block keeps $O(|C_t^{\text{free}}|^2)$ variables and the free–fixed block $O(|K_t|\,|C_t \setminus C_t^{\text{free}}|)$ coefficients; in the zones of Section 5.1 a round solves in 0.2–0.4 s on average and in at most 7.6 s (Appendix C.4), within the 10-second limit per stage. The programme is solved lexicographically in three stages, each fixing the previous optimum:

$$\max \sum_o y_o \;\succ\; \min v \;\succ\; \min\Big(\lambda_{\text{tt}} \sum_{(c,s)} \tau_s x_{cs} + \mu \sum_o \text{late}_o\Big), \tag{14}$$

subject to each order in at most one dispatched route, each courier in at most one, $y_o \le \sum_{(c,s)\ni o} x_{cs}$, and Eqs. (12) - (13). The complete formulation is given in Appendix B.4. Coverage is placed first because the alternative order (equity first) collapses from a cold start: when every courier has earned nothing, any assignment raises the projected Gini coefficient, so "assign nothing" minimises v in every round and the system never starts (coverage 0.000 on the pilot day from a cold start; 0.285 over the 27 runs of the ablation in Table C.14). With coverage first, the Gini excess decides *who* receives each route among the maximum-coverage assignments, which is the equity lever; $\alpha = 1$ switches the constraint off, which gives the coverage **ILP** on the same candidate pairs, and $\alpha \to 0$ minimises the projected absolute differences in every round. The equity-first variant is reported as an ablation.

Pay-BRKGA. **Pay-BRKGA** decides who is paid how much, not how routes are built. The chromosome of the coverage BRKGA is extended with one pay gene $g_c \in [0,1]$ per free courier, mapped to the multiplier $m_c = (1-\delta) + 2\delta g_c \in [1-\delta, 1+\delta]$ of whatever route the decoder assigns to c, and the offer is $\rho_{c,s} = \rho_0 \cdot |s| \cdot m_c$; under the affine tariff of Eq. (3) the multiplier would apply to ρ_1 only. The decoder is Algorithm 2. The fitness is the scalarised objective, in coverage-fraction units and to be minimised,

$$\Big(1 - \frac{\mathbb{E}[\text{covered}]}{|\mathcal{O}_t|}\Big) + \lambda_{\text{def}}\, F^{\text{proj}}_{\text{Deficit}} + \lambda_{\text{pay}}\Big(\frac{\sum \text{pay}}{\sum \text{tariff}} - 1\Big) + 10^{-3}\,\frac{\bar{\tau}}{20}, \tag{15}$$

where $\mathbb{E}[\text{covered}] = \sum_{(c,s)\in A} |s|\; p_{c,s}(\rho_{c,s})$ uses the acceptance model of Eq. (4), so that raising pay buys coverage only if the behavioural model says it does (in the auto-accept regime pay moves only F_{Deficit} and cost); $F^{\text{proj}}_{\text{Deficit}}$ is the fraction of free couriers whose projected hourly earnings after the round, assuming acceptance, remain below w^*; the third term is the relative pay excess over the tariff, the payment term of Eq. (6) normalised by the tariff; and the last term breaks ties by mean travel time $\bar{\tau}$. On these data $F^{\text{proj}}_{\text{Deficit}}$ is close to one for every chromosome, because no courier reaches w^* within a round (Section 6.3), so the term is inert at $\lambda_{\text{def}} = 0.5$ and the fitness trades, in practice, expected coverage against pay. The scalarisation is a weighted sum rather than an explicit Pareto set because a round yields one decision; sweeping λ_{pay} traces the cost-of-fairness

curve. We note that the mechanism **Pay-BRKGA** acts on is distance, not pay: under the flat tariff, a 2-km singleton pays 1,100 COP/km (acceptance about 0.64 at $\theta = 903$) while a 4-km one pays 550 COP/km (about 0.19), so the pay gene is the only lever that can make a long route acceptable in the manual regime.

Verification. The four pieces (BRKGA port, **BRKGA-Fair**, **MIP-Fair**, **Pay-BRKGA**) are covered by unit tests against exhaustive enumeration on a toy instance of three couriers and four orders; Appendix B.6 lists what each test checks. In summary, the three policies differ in the lever they move and in the solver, and share the candidate pairs and the two readings of the history.

5. Experimental setting

In the following, we outline the instances, the benchmark policies, the experimental design and the implementation details of our computational study.

5.1. Instances

We test our policies on 24 full operating days (00:00–24:00) exported by a leading on-demand delivery platform operating in Colombia [TODO: confirm whether the platform may be named]. Each day records every order (store, drop-off coordinates, placement, preparation and promised drop-off times) and every courier (vehicle, connection point, on and off times). Projecting the coordinates showed that the 24 days cover six cities: Bogotá, Medellín, Cali, Barranquilla, Bucaramanga and Pereira (four days each, except three for Cali and five for Bucaramanga). Full-city days range from 540 orders and 400 couriers (Pereira) to 59,000 orders and 16,800 couriers (Bogotá); fleet composition differs sharply, from 45% bicycles in Bogotá to 90% motorcycles in Pereira and Bucaramanga, which matters for the emissions diagnostic. The behavioural parameters come from the interviews and the survey described in Section 3.1; the remaining parameters are either platform practice or declared assumptions. The simulator, the policies, the instance loader and the batch scripts are available at https://github.com/Danielgiraldosb/mdrp-simulator [TODO: confirm whether the instance data can be shared; otherwise state the access conditions].

Layout. A full city day is not the problem a dispatcher solves, so the experimental unit is a *zone*: the orders (by drop-off) and couriers (by connection point) within 3 km of the instance's store centroid, over the full day. Three kilometres keeps Pereira above a few hundred orders, so that end-of-day Gini coefficients are not noise, and Bogotá below the scale at which the per-round MIP exceeds the budget. Each day is projected on a plane centred on the store centroid, discarding a few points hundreds of kilometres away. Table C.6 in Appendix C.1 lists the 24 instances; zone sizes range from 342 to 10,215 orders (mean 3,612), and Figure C.3 in Appendix C.1 shows the zone of instance 0 in Bogotá with its stores, drop-offs and courier connection points. Two consequences of the zone follow the comparisons throughout: the couriers who connect inside the circle serve the

whole city in reality while only the orders dropped inside it are kept, so the fleet is over-supplied relative to the field (0.36 orders per connected courier-hour in the zones, against about 1.2 implied by the survey median of 2,622 COP/h at the flat tariff), and about a fifth of the orders cannot be delivered on time by any courier of the zone, which bounds coverage near 0.79 for every policy; both are discussed in Section 8.

Couriers and travel. Travel time is Euclidean distance times a circuity factor of $\kappa = 1.3$, standard for an urban street network relative to the straight line (1.0 in a sensitivity run), divided by the vehicle speed: 15 km/h for bicycles, 25 for motorcycles, 5 for walking and 20 for cars, declared as assumptions of the same order as the 320 m/min of the public MDRP instances (Reyes et al. 2018). Couriers connect and disconnect at the recorded on and off times; repositioning of idle couriers is switched off because most surveyed couriers wait in place. A courier decides on an offer after a normally distributed delay with mean 16 s and standard deviation 8 s, truncated at 3 s.

Orders and service. In-store service takes 20 minutes with a purchased order (60 minutes in a stress run) and drop-off service 4 minutes, the default of the public instances. The hard deadline is the promised drop-off time plus $\Delta^{\max} = 30$ minutes, the slack the interviews describe as tolerated. Route capacity is $Q = 3$. Customer cancellations are switched off because there is no field evidence on them.

Tariff, acceptance and wage. The flat tariff pays $\rho_0 = 2{,}200$ COP per order (survey, $n = 132$); the pay band of **Pay-BRKGA** is $\delta = 0.30$. The logit of Eq. (4) uses $\theta = 903$ COP/km and $b = 2.89$, the model estimated on the 32 interview decisions under the 2020 manual regime. Figure C.4 in Appendix C.1 shows the curve with its interval and the two regimes; the full estimation is in the online supplement. The statutory minimum hourly wage is $w^* = 4{,}367.91$ COP (2021 monthly minimum divided by 48×4.33 hours). Emission factors are 0.114 kg CO_2e/km for motorcycles and 0.167 for cars (Department for Energy Security and Net Zero and Department for Environment, Food & Rural Affairs (UK) 2023), zero for bicycles and walking; local factors by vehicle type do not exist. Table C.5 in Appendix C.1 summarises the parameters with their sources and, for the behavioural ones, the evidence grade.

5.2. Benchmark policies

We compare the three proposed policies against four benchmarks designed to isolate the value of the equity history, of the exact constraint and of the pay lever.

- **Greedy**: Courier-centric. The free couriers are visited in increasing order of the orders they have delivered so far, and each takes the best feasible route (a bundle when one is feasible, otherwise the nearest order) at its nearest store with pending orders. It is our implementation of the platform's practice as the couriers describe it, "the free couriers pick work in turn"; the least-served-first order is our reading of "in turn", since the platform does not disclose

its queue rule, and it makes **Greedy** read one number of $\mathcal{H}_t$, the order count, as a priority rule. It is the reference for all paired comparisons, and its end-of-day Gini coefficient is the historical anchor for α.

- **ILP**: A per-round integer programme over K_t that maximises the number of orders covered, with a secondary term for travel time. It is the exact per-window coverage dispatcher and the model that **MIP-Fair** extends; the two differ in the candidate prefilter ($k = 25$ against $k = 8$, Section 5.5), so the comparison that isolates the Gini constraint is **MIP-Fair** at $\alpha = 1$ against $\alpha < 1$ (Section 6.1).
- **BRKGA**: The coverage BRKGA of Section 4 with $\gamma = 0$ and the tariff pay. Comparing it with **BRKGA-Fair** isolates the deficit term, and with **Pay-BRKGA** the pay gene.
- **Hierarchical**: A per-round lexicographic policy that first maximises coverage, then minimises lateness, then balances the pay offered among the free couriers of the round, then minimises cost. It is fair within each window and ignores $\mathcal{H}_t$; it is the empirical counterpart of the example in Appendix B.2, and comparing it with **BRKGA-Fair** and **MIP-Fair** isolates the value of carrying the history in the state.

5.3. Experimental design

Table C.7 in Appendix C.1 lists the factors and levels. Eighteen policy configurations (four benchmarks, four weights γ of **BRKGA-Fair**, six thresholds of **MIP-Fair** and four pay weights of **Pay-BRKGA**) are crossed with the two acceptance regimes, the 24 instances and ten seeds. Seeds are common random numbers (1 to 10): every comparison between policies is paired by (instance, seed), and the seed drives the acceptance draws, the decision delays and the BRKGA. The α grid is centred on the inequality band of Section 3.1.

The study consists of four experiments. **X0**, the pilot, runs one instance per city for the seven policies at default settings under the manual regime with one seed (42 runs) to measure run times and the corporate Gini coefficient. **X1**, the main experiment, runs the 18 configurations for both regimes on the 24 instances with ten seeds (8,640 runs) and produces the frontiers, the cost of the guarantee, the compliance and the regime loss. **X2**, the threshold experiment, re-runs five selected configurations (**Greedy**, **Hierarchical**, **BRKGA-Fair** with $\gamma = 40$ and $k_d = 25$, **MIP-Fair** with $\alpha = 0.30$ and $k_d = 25$, **Pay-BRKGA** with $\lambda_{\text{pay}} = 0.5$) at the bounds of the threshold's confidence interval under the manual regime, the only one in which the threshold is read, with five seeds (1,200 runs, 1,197 completed). **X3** evaluates, with five seeds, the second-generation variants of Section 6.1 (exposure basis, deficit-aware prefilter with $k_d \in \{8, 25\}$ and coverage slack; eight configurations under both regimes, 1,920 runs) and a supplement of eleven ablations (decoder score, population size, scope and basis of the Gini constraint, equity-first priority, prefilter $k = 25$, coverage slack, and $k_d = 50$; manual regime except $k_d = 50$, 1,453 runs completed; Appendix C.6). In total 13,255 runs and 1,239 CPU-hours: a run took 2.9 minutes on average in X1, 7.1 in the second generation of X3 and in X2, and 18.7 in the ablations.

5.4. Metrics and analysis

For every run we record the two components of the cost of Eq. (5) separately, coverage (fraction of orders delivered) and mean and total lateness, the number of late orders, click-to-door mean and 90th percentile, the end-of-day F_{Gini}, F_{MaxMin}, F_{Orders} and F_{Deficit}, total payment, the number of active couriers, the mean bundle size, the mean and maximum solve time per round, and the wall-clock time; emissions follow from the per-courier distances by vehicle. The reported Gini coefficient is one and the same for all policies, Eq. (7) over the eligible couriers, computed ex post from the per-courier table of every run; the active-courier statistic $F^{\text{act}}_{\text{Gini}}$ is reported beside it, and the scope used inside a policy affects what it optimises, not what is measured. To compare the policies, we compute for each policy π and instance i the average $\bar{y}^{\pi}_{i}$ of a metric over the ten seeds and report the mean over the 24 instances together with its standard error, $\text{SE} = \text{SD}/\sqrt{N}$ with $N = 24$ and SD the sample standard deviation across instances. The cost of the guarantee is the paired difference $\bar{y}^{\pi}_{i} - \bar{y}^{\mathbf{Greedy}}_{i}$ with a 95% percentile bootstrap interval (2,000 resamples) over the 24 instances (instance as block, seeds averaged within); the paired differences between a variant and its parent in Section 6.1 and in Appendix C.3 and Appendix C.6 are over the (instance, seed) pairs with the normal approximation $\pm 1.96\,\text{SE}$. The share of the gap to **Greedy** closed by a variant is $1-(\bar{F}^{\text{var}}-\bar{F}^{\mathbf{Greedy}})/(\bar{F}^{\text{par}}-\bar{F}^{\mathbf{Greedy}})$, a ratio of means over instances on the same seeds, so that it can be recomputed from the tables. Compliance is the fraction of (instance, seed) pairs, which we call days for short, with $F_{\text{Gini}} \le \alpha$ by policy and α, computed ex post, so the grid of α can be extended without new runs. Under the flat tariff F_{Orders} coincides with $F^{\text{act}}_{\text{Gini}}$ for every policy that pays the tariff, so it is not reported separately; F_{MaxMin}, click-to-door and the solve time per round are in Appendix C.4. The regime loss is the paired difference manual minus auto-accept. Frontiers are the mean over (instance, seed) of $(F_{\text{Gini}}, \text{coverage})$ and $(F_{\text{Gini}}, \text{lateness})$ at each level of the swept parameter, with the non-dominated envelope and the knee selected by the maximum distance from the chord between the extremes, applied identically to the three policies.

5.5. Implementation details

The parameters of the policies were tuned in preliminary experiments. The BRKGA uses a population of 10 and 10 generations per round, elite and mutant fractions of 20%, elite bias 0.7 and $K = 3$ couriers per order in the decoder; a population of 20 with 20 generations is an ablation. The candidate prefilter keeps the $k = 25$ nearest available couriers per route for all policies except **MIP-Fair**, which uses $k = 8$ for budget reasons: with $k = 25$ a two-hour window of Bogotá did not finish within seven minutes, while with $k = 8$ it takes 4.5 minutes and covers 0.443 against 0.437 for **Greedy**; $k = 25$ is an ablation. **Pay-BRKGA** uses $\lambda_{\text{def}} = 0.5$. Mixed-integer programmes are solved with a limit of 10 seconds per round, and each run has a timeout of 90 minutes in X1, 150 in X2 and in the second generation of X3, and 240 in the ablations of X3. All experiments were conducted on a computer equipped with an Intel Core i9-14900K processor with turbo disabled, running Windows 10, with 12 parallel processes; the CPU times reported are wall-clock minutes

per run under that load, so ratios between policies are comparable and absolute values conservative. The implementation was carried out in Python 3.10, and Gurobi 11.0 was used for the **ILP**, **MIP-Fair** and **Hierarchical** policies.

6. Computational study

In this section, we present our computational study. We first compare the policies with respect to coverage, punctuality and the end-of-day Gini coefficient, and report whether they comply with the guarantee. Then, we perform an in-depth analysis of the equity history: how the Gini coefficient evolves within the day under the per-window policy and under the policies that read $\mathcal{H}_t$, and how the three policies change who receives which route and what it pays. Finally, we examine how policy performance changes across operational settings: the acceptance regime, the behavioural threshold, the tolerated inequality and the fleet composition of the six cities.

6.1. Solution quality and compliance

To assess the solution quality of the policies, we use the **Greedy** dispatcher as the base benchmark, since it reproduces the platform's practice, and report for each policy its best configuration per regime (the configuration that satisfies the band criterion $\alpha = 0.30$ on most instances at the least coverage cost, or, where none does, the one with the lowest Gini coefficient). Table 2 reports, by regime, the mean over the 24 instances with its standard error; the best value per column and regime is in bold (higher for coverage and active eligible, lower for the Gini coefficients, lateness, the minimum-wage deficit and CPU time; payment is not ranked, since lower is better for the platform and worse for the couriers). The selection is made on the same runs on which the policies are then compared, which can only favour the proposed policies; the full grid traces the frontiers of Figures 1 and C.5.

Three facts organise the table. First, under the auto-accept regime coverage is identical for the 18 configurations, 0.786 ± 0.013: the orders that are not delivered (21%) cannot be delivered on time by any courier of the zone, so the fleet is not binding and the policies differ only in who delivers, when, and at what pay. Under the manual regime rejection opens a spread in coverage, from 0.738 (**ILP**) to 0.770 (**BRKGA** and **BRKGA-Fair**). Second, **Greedy** is dominated on the service side: every other policy except **ILP** and **Pay-BRKGA** delivers at least as much with less lateness (-0.9 to -1.8 minutes, paired intervals excluding zero). **ILP** covers least under manual acceptance, which is consistent with a dispatcher that maximises coverage as if every offer were accepted and therefore offers the long routes that the logit rejects most often. Third, and this is the central result of the paper, **Greedy** is the most equitable policy by a wide margin on the statistic of Eq. (7): 0.301 under the manual regime and 0.296 under auto-accept, against 0.501–0.566 for every other policy, the three proposed ones included. The paired differences with respect to **Greedy** are between $+0.205$ and $+0.270$, every 95% interval at least 0.17 from zero, and the ranking holds in each of the six cities.

Policy	Coverage mean	Coverage SE	End-of-day Gini F_{Gini} mean	End-of-day Gini F_{Gini} SE	Active eligible	Gini $F^{\text{act}}_{\text{Gini}}$	Lateness (min)	Min.-wage deficit F_{Deficit}	Payment (% of tariff)	CPU time (min/run)
Manual regime										
Greedy	0.758	0.013	**0.301**	0.008	**0.97**	0.509	19.9	0.981	100	2.2
ILP	0.738	0.013	0.521	0.015	0.73	**0.414**	19.4	0.993	100	**1.9**
Hierarchical	0.769	0.013	0.527	0.011	0.77	0.530	**18.1**	0.981	100	3.2
BRKGA	0.770	0.013	0.565	0.014	0.70	0.488	18.5	0.984	100	2.3
BRKGA-Fair ($\gamma = 40$)	0.770	0.013	0.534	0.015	0.75	0.504	18.6	0.985	100	2.4
MIP-Fair ($\alpha = 0.30$)	0.764	0.013	0.544	0.013	0.72	0.460	19.0	0.988	100	4.5
Pay-BRKGA ($\lambda_{\text{pay}} = 0.5$)	0.753	0.013	0.546	0.013	0.71	0.457	18.8	0.991	88	2.8
Auto-accept regime										
Greedy	0.786	0.013	**0.296**	0.008	**0.97**	0.586	19.0	**0.973**	100	1.8
ILP	0.786	0.013	0.501	0.017	0.75	**0.395**	18.8	0.993	100	**1.4**
Hierarchical	0.786	0.013	0.515	0.010	0.79	0.570	**17.9**	0.977	100	2.8
BRKGA	0.786	0.013	0.566	0.014	0.70	0.509	18.1	0.982	100	2.0
BRKGA-Fair ($\gamma = 40$)	0.786	0.013	0.536	0.015	0.76	0.516	18.2	0.983	100	2.1
MIP-Fair ($\alpha = 0.30$)	0.786	0.013	0.534	0.015	0.73	0.450	18.8	0.988	100	3.2
Pay-BRKGA ($\lambda_{\text{pay}} = 0.5$)	0.786	0.013	0.535	0.013	0.73	0.456	18.3	0.992	80	2.2

Table 2: Comparison of the policies by acceptance regime: mean over the 24 instances (seeds averaged within each instance) and standard error. F_{Gini} is the Gini coefficient of hourly earnings over the eligible couriers (connected ≥ 1 h, idle ones at zero), "active eligible" the share of eligible couriers that delivered at least one order, and $F^{\text{act}}_{\text{Gini}}$ the same Gini coefficient restricted to them. Payment is relative to the flat tariff of the orders delivered

The mechanism is who gets work. **Greedy** is courier-centric: every free courier takes its best route, the least-served courier first, so 97% of the eligible couriers deliver at least one order during the day. The exact **ILP**, **MIP-Fair**, the BRKGA family and **Hierarchical** are order-centric: each round assigns the orders to the best-placed couriers, the same couriers tend to be best placed round after round, and between 21% and 31% of the eligible couriers end the day without a single delivery and with $w_c = 0$. The first quartile of hourly earnings among eligible couriers is 481 COP under **Greedy** and between 86 and 146 COP under the other policies. Table C.13 in Appendix C.5 shows how this inverts the statistic that the fairness literature, and the first version of this study, computed over the active couriers: $F^{\text{act}}_{\text{Gini}}$ is lowest for **ILP** (0.414), then **Pay-BRKGA** and **MIP-Fair** (0.457, 0.460), and highest for **Greedy** (0.509) and **Hierarchical** (0.530). Two manoeuvres produce the inversion. Idle couriers leave the active population, so concentrating work on fewer couriers lowers $F^{\text{act}}_{\text{Gini}}$ while raising F_{Gini}; and the 7% of couriers who connect for less than one hour, to whom **Greedy** dispatches because they are free and nearby, enter $F^{\text{act}}_{\text{Gini}}$ with hourly earnings of up to 110,000 COP (one delivery of 2,200 COP within a few minutes of connecting) and raise it, whereas Eq. (7) excludes them. Restricted to the eligible couriers that were active, **Greedy** is still the most equitable (0.28 against 0.35–0.39), so the first manoeuvre is the larger one.

The equity history did not prevent this because of what the three policies read in it. The deficit of **BRKGA-Fair** and the projected w_c of **MIP-Fair** are computed per *busy* hour, a productivity rate: a courier with one ten-minute delivery and a courier with twenty have nearly the same value, and only a courier who has never delivered has a deficit. The signal therefore separates couriers with and without work but not couriers with little and much, and it competes with coverage, which has lexicographic priority in **MIP-Fair** and is the fitness of the BRKGA population. The deficit bias does act in the right direction: **BRKGA-Fair** raises the active share of eligible couriers from

0.70 (**BRKGA**) to 0.75 and lowers F_{Gini} from 0.565 to 0.534–0.547 as γ grows from 5 to 40, in both regimes, and **MIP-Fair** activates more couriers than **BRKGA**; but the effect is a fraction of the gap to **Greedy**. The α of **MIP-Fair** has no measurable effect (0.544–0.545 under the manual regime, including $\alpha = 1.00$, which switches the constraint off), because with coverage first and $k = 8$ candidates the coverage-optimal assignments of a round leave the Gini block almost nothing to choose. The second-generation variants that correct this are evaluated after Table C.13. The minimum-wage deficit is between 0.973 and 0.993 under every policy, consistent with the 92% of surveyed couriers below the statutory minimum. On CPU time, **ILP** is the cheapest policy and **MIP-Fair** the most expensive (4.5 minutes per day on average, 22 at most on a Bogotá instance); the three proposed policies cost between 1.1 and 2.1 times **Greedy**.

Second-generation variants. Two changes follow from this diagnosis. First, the history $\mathcal{H}_t$ that the three policies read is the *exposure* basis of Eq. (7), earnings per hour connected so far over the eligible couriers, so that a courier with few deliveries over a long connection carries a large deficit instead of looking productive. Second, the deficit-aware prefilter adds to the candidate pairs of each route, after its k nearest available couriers, the k_d available couriers with the largest deficit, the nearest among ties, so that the idle couriers, whose deficit is one, are at least offered to the solver. Table C.9 in Appendix C.3 reports the variants on the five seeds shared with X1 against their first-generation parents and **Greedy**, and Table C.10 the paired differences with their intervals, the share of the gap closed and the compliance of every variant. The basis alone is almost inert: it moves F_{Gini} by 0.004–0.009 (**MIP-Fair**) and 0.023–0.025 (**BRKGA-Fair**) and closes 2–10% of the gap to **Greedy**, because an idle courier who is not a candidate cannot be chosen however large its deficit. The prefilter is the lever, and its effect grows with k_d: with $k_d = 8$ the gap closes by 23% (**BRKGA-Fair**) and 33–45% (**MIP-Fair**); with $k_d = 25$ by 42–50% (**BRKGA-Fair**, $\gamma = 40$) and 45–56% (**MIP-Fair**), to $F_{\text{Gini}} = 0.435$–0.437 under the manual regime and 0.401–0.416 under auto-accept, with 85–91% of the eligible couriers active. Every paired interval excludes zero and no (instance, seed) pair is worse than its parent at $k_d = 25$ (Table C.10). The service cost is nil: coverage changes by -0.001 to $+0.003$ with respect to the parent (the **MIP-Fair** variants gain slightly, because the extra candidates also serve coverage), lateness rises by 0.1–0.3 minutes, and the regime loss is unchanged. The coverage slack ε of **MIP-Fair** does nothing ($\varepsilon = 0.05$ against 0: F_{Gini} within 0.002 and coverage within 0.001), which confirms that what bound the Gini block was the candidate set and not the lexicographic priority. **Pay-BRKGA** responds least to the prefilter (8–11% of the gap over the two regimes) because its fitness has no equity term: only the prefilter reads the deficit, and the decoder then scores on coverage alone. The price of k_d is CPU time: **BRKGA-Fair** with $k_d = 25$ costs 4.3–4.9 minutes per day on average and 25 at most, twice its parent; **MIP-Fair** with $k_d = 25$ costs 11–24 minutes on average and up to 110 on a Bogotá day under the manual regime, five times its parent, for the same end-of-day Gini, so the heuristic is the practical choice at this scale. Half of the gap to **Greedy** remains, and the ablation with $k_d = 50$

(Table C.14 in Appendix C.6) shows that for **BRKGA-Fair** it is a matter of candidates, since F_{Gini} keeps falling to 0.404 under the manual regime and 0.352 under auto-accept at unchanged coverage and 1.3 times the CPU time, whereas for **MIP-Fair** the curve flattens (0.412 and 0.398) while the CPU time grows to 51 minutes per day: for the exact round model the one-route-per-courier structure and the solver limit bind before the candidate set does.

Table C.12 in Appendix C.5 reports the compliance with the guarantee, the fraction of (instance, seed) pairs with $F_{\text{Gini}} \leq \alpha$, by policy and α. **Greedy** meets $\alpha = 0.30$, the inside-the-ladder anchor of Section 3.1, on half of the days, 0.35 on nine days in ten and 0.40 on every day under auto-accept. No other first-generation configuration meets $\alpha \leq 0.40$ on more than 4% of the days under either regime (**ILP** under auto-accept); at 0.45 the best of them (**ILP**) reaches 18% of the days under the manual regime and 46% under auto-accept, and at 0.50 the proposed policies reach 35–50%. The lower half of the α grid is therefore within reach only of the dispatcher that spreads the work, and the parameter of each proposed policy moves compliance by a few points at most. The second generation changes the lower half of the grid (last block of the table): with $k_d = 25$, **BRKGA-Fair** and **MIP-Fair** meet $\alpha = 0.40$ on 40–49% of the days under the manual regime and 56–63% under auto-accept, 0.45 on 61–62% and 67–83%, and 0.50 on 72–82% and 81–96%, and still on at most 2% at 0.30. We therefore take $\alpha^* = 0.45$ as the operating point of the threshold experiment of Section 6.4: not a value derived from the couriers' or a regulator's perspective, but the tightest level of the grid that an order-centric policy currently meets on most days under both regimes at no coverage cost (Table C.16).

Figures 1 and C.5 (the latter in Appendix C.4) show the frontiers traced by the three policies in their own parameter: α for **MIP-Fair**, γ for **BRKGA-Fair** and λ_{pay} for **Pay-BRKGA**. The x-axis is the end-of-day Gini coefficient of Eq. (7), the y-axis coverage (Figure 1) and mean lateness (Figure C.5), each point is a configuration averaged over the 24 instances and the seeds available, and the benchmarks appear as single points. The three frontiers occupy a narrow band on the right of the figures, between 0.53 and 0.57, and **Greedy** stands alone on the left at 0.30 with less coverage and more lateness than any of them: no configuration of the three policies trades service for equity along its own parameter at a scale comparable to the gap between the two groups. **BRKGA-Fair** is the only sweep with a monotone direction, moving 0.03 to the left as γ grows at constant coverage; **MIP-Fair** collapses to a point in α; **Pay-BRKGA** moves along coverage (manual) and lateness rather than along the Gini axis as λ_{pay} grows. The second-generation variants (hollow markers, five seeds) populate the region between the two groups along k_d rather than along the original parameters: from the parent at 0.53–0.54 to 0.47–0.48 at $k_d = 8$ and 0.44 at $k_d = 25$ under the manual regime (0.40–0.42 under auto-accept), at constant coverage and with lateness within 0.3 minutes of the parent. The gap that the first-generation parameters could not close is halved by a change in the candidate set, not in the objective; **Pay-BRKGA** moves least (0.02).

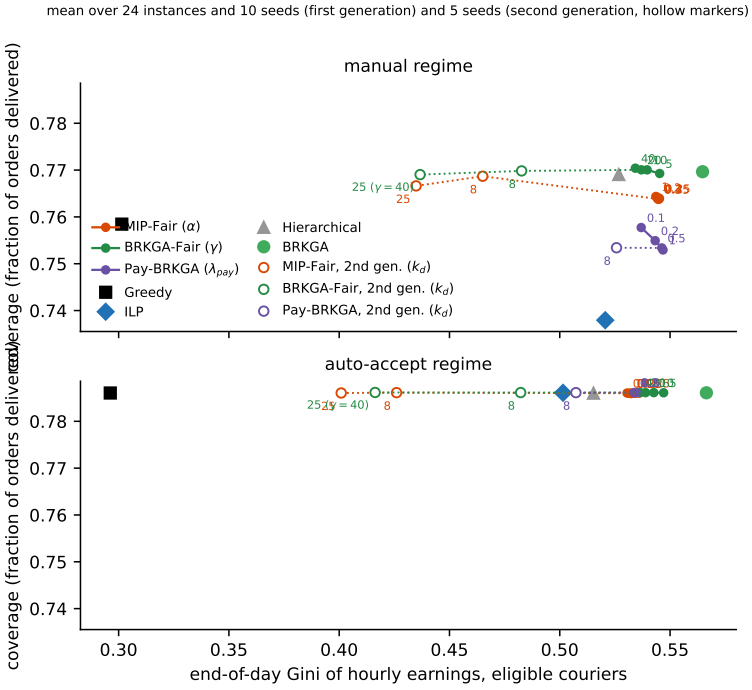


Figure 1: Coverage against the end-of-day Gini coefficient by policy, manual (top) and auto-accept (bottom) regime; solid paths: first-generation sweeps in their own parameter; hollow markers and dotted paths: second-generation variants along k_d (five seeds)

6.2. The equity history within the day

One of our main modelling contributions is the equity history in the state. In this section, we analyse how the Gini coefficient evolves within the day under the policies that read $\mathcal{H}_t$ and under the policy that does not. Figure 2 shows the trajectory of the Gini coefficient of hourly earnings so far, computed every round over the eligible couriers connected up to that round, idle ones at zero, for **Greedy**, **Hierarchical**, **MIP-Fair** ($\alpha = 0.30$) and the second-generation **BRKGA-Fair** of Section 6.1 ($\gamma = 40$, deficit-aware prefilter with $k_d = 25$) on one Bogotá day (instance 0, manual regime, seed 1: 10,215 orders, 2,650 couriers in the zone, of whom 2,560 are eligible). The x-axis is the time of day, the y-axis the Gini coefficient, recomputed every ten minutes from the delivery log; the dashed line marks $\alpha = 0.40$, the upper bound of the inequality band. The left panel is the statistic of the paper; the right panel is the same coefficient over the couriers that have delivered at least once so far, the population on which the first-generation policies read their history. The four trajectories start near 1: at 08:00, 525 eligible couriers are connected and about forty have delivered, so almost everyone is at zero, and the coefficient falls as orders arrive and the fleet is put to work. The policies separate between 10:00 and 16:00, the hours in which most of the fleet connects. **Greedy** puts the new couriers to work as they arrive and its trajectory falls from 0.79 at 10:00 to 0.41 at 14:00, crosses $\alpha = 0.40$ at 14:12 and settles at 0.23 from 16:00 on. **Hierarchical** and **MIP-Fair** keep assigning to the couriers already in motion, so their trajectories

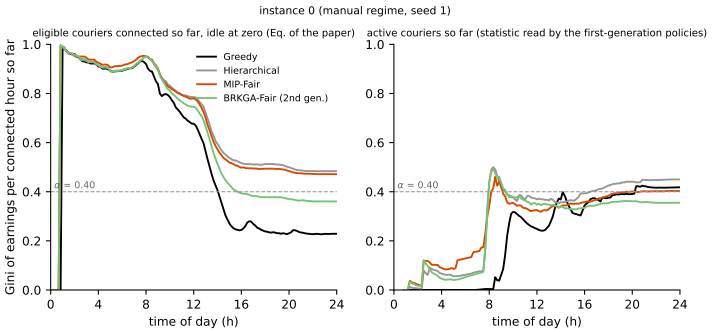


Figure 2: Intraday trajectory of the Gini coefficient of earnings per connected hour on one Bogotá day (instance 0, manual regime, seed 1), recomputed every ten minutes: over the eligible couriers connected so far with idle ones at zero (left, the statistic of Eq. (7)) and over the couriers with at least one delivery so far (right). Dashed line: $\alpha = 0.40$

fall more slowly (0.59 and 0.57 at 14:00) and never reach α: they end the day at 0.48 and 0.47 with 15–16% of the eligible couriers idle. The right panel shows why neither policy reacted: on the active couriers both are at 0.33–0.39 all afternoon, below α and below **Greedy**, so the statistics they control report compliance while the end-of-day statistic of Eq. (9) is 0.2 above the guarantee, because the couriers who make the difference are not in their population. The second-generation **BRKGA-Fair**, which admits the idle eligible couriers with the largest deficit as candidates in every round, follows the order-centric trajectory until noon and then bends towards **Greedy**: 0.51 at 14:00, below α at 15:48, and 0.36 at the end of the day with 97% of the eligible couriers active and coverage 0.780, the highest of the four. The two statistics thus move in different directions because they are computed on different populations, and the end-of-day one is decided by the hours in which the fleet grows.

6.3. Decision making

We next analyse how the three policies change the dispatching decision compared to **Greedy**: who receives which route, and what it pays. Figure C.6 in Appendix C.5 compares the distribution of end-of-day hourly earnings over the eligible couriers under **Greedy** and under each policy at its best configuration, pooled over the 24 instances of one seed (23,944 couriers, manual regime); the x-axis lists the policies, the y-axis the earnings per connected hour, and the share of couriers who end the day idle is printed under each violin. The statutory minimum $w^* = 4{,}368$ COP/h lies above the 99th percentile of every distribution, which is the content of $F_{\text{Deficit}} \geq 0.97$. The policies do not compress the distribution of **Greedy**; they stretch it at both ends. **Greedy** leaves 3% of the eligible couriers idle and its interquartile range is 519–876 COP/h around a median of 685; the six other policies leave 19–25% idle, their first quartile is between 0 (**BRKGA**) and 205 COP/h, their median is similar (547–621) and their third quartile is higher (1,066–1,206 COP/h): the orders that **Greedy** spreads over the couriers who happen to be free go, under an order-centric policy, to the couriers who are best placed, who are then best placed again. The reduction of the Gini coefficient

of Eq. (7) under **Greedy** therefore comes from below (fewer idle couriers) and from above (fewer high earners) at once, at the same total pay: the mean is 801 COP/h under **Greedy** and 708–796 under the policies that pay the tariff. **Pay-BRKGA** shifts the whole distribution down (mean 689, median 547) because it pays 12% less, and still leaves 23% idle. No courier crosses w^* under **Pay-BRKGA** either. Bundling plays no role in these differences: the mean bundle size is 1.03 orders per route under **Greedy** and the BRKGA-type policies, 1.01 under **MIP-Fair** and 1.00 under the **ILP** and **Hierarchical** policies. For **Pay-BRKGA** the pay goes where acceptance is at stake. The total payment is 88% of the tariff under the manual regime and 80% under auto-accept (Table 2), while the median pay per kilometre received by an active courier is 929 COP/km against 816 under **Greedy** in the manual regime (789 against 669 under auto-accept): the multiplier raises the pay of the long routes that would otherwise be rejected and lowers that of the short ones that would not, so the bill falls while the pay per kilometre of what is accepted stays above the threshold. In the auto-accept regime, where nothing is rejected, the same rule only lowers the bill and leaves F_{Deficit} unchanged.

6.4. Sensitivity analysis across operational settings

Finally, we study how different operational conditions affect the performance and relative advantages of the policies. We vary one factor at a time around the base case (manual regime, $\theta = 903$ COP/km, the α grid) and report, for the benchmark dispatchers and the proposed policies at their selected configurations, the relative difference over the base-case value of the policy, following the one-factor-at-a-time design. Table 3 reports the results for coverage; the regime and α columns come from X1, the θ columns from X2, which re-runs five configurations at the ends of the 95% interval of the logit. Every cell is relative to the same configuration's base-case value on the seeds of its experiment (ten in X1, five in X2), and a dash marks a parameter the policy does not have or a configuration not run in X2. Circuity is not in the table: in the pilot, with one seed and one instance per city, **Greedy** covered 6% more at circuity 1.0 than at 1.3.

	Relative difference over the base-case setting (in %)							
Policy\Setting	**Regime**		**Threshold** θ (COP/km)			**Tolerated inequality** α		
	manual	**auto**	**615**	**903**	**1,242**	**0.20**	**0.30**	**0.40**
Greedy	0.0	+3.6	+3.2	0.0	−9.0	–	–	–
Hierarchical	0.0	+2.2	+2.0	0.0	−5.5	–	–	–
MIP-Fair ($\alpha = 0.30$)	0.0	+2.9	–	0.0	–	0.0	0.0	0.0
BRKGA-Fair ($\gamma = 40$, $k_d = 25$)	0.0	+2.2	+2.0	0.0	−5.7	–	–	–
MIP-Fair ($\alpha = 0.30$, $k_d = 25$)	0.0	+2.5	+2.3	0.0	−7.0	–	–	–
Pay-BRKGA ($\lambda_{\text{pay}} = 0.5$)	0.0	+4.3	+3.6	0.0	−7.3	–	–	–

Table 3: Sensitivity analysis across operational settings: relative difference of coverage over the base case (manual regime, $\theta = 903$, $\alpha = 0.30$). Regime and α from X1 (ten seeds); θ from X2 (five seeds, the five configurations of the threshold experiment); a dash marks a parameter the policy does not have or a configuration not run in X2. The companion for the Gini coefficient is Table C.15 in Appendix C.7

Acceptance regime. Figure C.7 in Appendix C.7 shows the regime loss, the paired difference manual minus auto-accept by instance, for the seven configurations of Table 2. In coverage the loss is between 1.6 points (**BRKGA**, **BRKGA-Fair**) and 4.8 points (**ILP**), with 2.8 for **Greedy**, 2.2 for **MIP-Fair** and 3.3 for **Pay-BRKGA**; every interval excludes zero, and the loss of **ILP** alone exceeds the whole spread of coverage between policies within the manual regime (3.2 points). The policies that re-optimise the round after a rejection lose least. **Pay-BRKGA** loses 1.7 points more than **BRKGA**: its pay cut lowers the acceptance probability of the long routes on which the multiplier would have to rise, so in the manual regime the pay lever is paid for in coverage, whereas under auto-accept the same lever cuts the bill by 20% at no coverage cost. The regime moves the end-of-day Gini coefficient by at most 0.02 (+0.020 for **ILP**, −0.002 for the BRKGA family), an order of magnitude less than the gap between **Greedy** and the other policies, so the ranking of Table 2 is the same under both regimes. With a flat tariff and a threshold of 903 COP/km, a 4-km singleton is accepted with probability 0.19, so coverage under manual acceptance is bounded by behaviour rather than by dispatching, and auto-accept removes that bound by moving the courier's decision to connection time: the regime matters more than the policy for coverage, and the policy matters more than the regime for equity.

Behavioural threshold. The threshold θ is a grade-C parameter estimated from 32 decisions, so X2 re-runs the five configurations of Section 5.3 at the ends of its 95% interval, $\theta = 615$ and 1,242 COP/km, paired with the same seeds at $\theta = 903$ (three **MIP-Fair** runs on the largest Bogotá day exceeded the timeout at $\theta = 1{,}242$ and their pairs are dropped from the comparison). With a flat tariff of 2,200 COP per order, a single order is accepted with probability one half at 3.6 km when $\theta = 615$, at 2.4 km when $\theta = 903$ and at 1.8 km when $\theta = 1{,}242$, so the response of coverage is asymmetric: the lower threshold adds 1.5–2.7 points (+2.0 to +3.6%) and the upper one removes 4.2–6.9 points (−5.5 to −9.0%). The policies that re-optimise the round after a rejection lose least (**Hierarchical** −5.5%, **BRKGA-Fair** −5.7%) and **Greedy** loses most (−9.0%), so the coverage ranking changes at the upper end of the interval: **Greedy**, fourth of five at $\theta = 903$, is last at 1,242, 3.7 points below **Hierarchical**, and **Pay-BRKGA**, last at 903, overtakes it because its multiplier keeps the long routes above the threshold. The end-of-day Gini coefficient is less sensitive than coverage, as expected from a statistic normalised by connected time: it moves by −0.7 to −2.8% at 615 and by +2.2 to +3.8% at 1,242, every paired interval excluding zero but no difference larger than 0.017, and the fairness ranking is the same at the three thresholds (**Greedy** 0.30–0.31, the two second-generation policies 0.42–0.45, **Hierarchical** 0.52–0.54, **Pay-BRKGA** 0.54–0.56). Compliance with $\alpha = 0.45$ is 63%, 62% and 52% of the days for **BRKGA-Fair** and 73%, 61% and 50% for **MIP-Fair** at 615, 903 and 1,242: the recommendation of Table C.16 holds inside the interval, with ten points of compliance lost at its upper end.

Tolerated inequality and the choice of α. Three complementary criteria are used to choose α (Table C.16 in Appendix C.7): A, the historical anchor, the end-of-day Gini coefficient of **Greedy**,

0.30 on the ten seeds of X1 (0.27–0.32 by city); B, the inequality band, whose conventional bounds read a Gini coefficient below about 0.30 as low and above about 0.40 as high, placed with the survey anchors of 0.26 inside the bonus ladder and 0.39 outside it (Gini coefficients of daily income among active respondents, comparable with the active-courier statistic rather than with Eq. (7)), so $\alpha = 0.30$; and C, the Pareto knee, the point of maximum distance from the chord between the extremes of the coverage–Gini frontier. Criteria A and B coincide at 0.30 on these data, because **Greedy** is itself inside the low-inequality band, and the only policy that meets it is **Greedy**, on half of the days. Criterion C does not apply: coverage does not fall along either sweep, so there is no knee to find. The interval is therefore $[0.30, 0.45]$: 0.30 is the status quo, which the order-centric policies do not reach on any day; 0.40, the upper bound of the band, is met on 40–63% of the days by the second generation at no coverage cost; and 0.45, above the band of criterion B, is the smallest value that the order-centric policies meet on most days under both regimes. The recommendation is therefore conditional: a platform that values the coverage and lateness gain of the order-centric policies can operate at $\alpha^* = 0.45$ with **BRKGA-Fair** ($\gamma = 40$, $k_d = 25$), which costs 4–5 minutes of CPU per day against 11–24 for **MIP-Fair** at the same compliance; a platform that holds to $\alpha \leq 0.35$ should keep the courier-centric rule, which meets it on nine days in ten.

Fleet composition and emissions. Fairness in hourly earnings is neutral to vehicle type by construction; any emissions effect comes from which couriers receive distance, so the diagnostic is informative precisely where fleets are mixed. Table C.17 and Figure C.8 in Appendix C.7 report the diagnostic from the per-courier distances and the emission factors of Table C.5. Fleet composition sets the level: a delivered order costs 0.13–0.17 kg CO_2e in Bogotá, where bicycles and walkers carry a quarter of the distance, and 0.26–0.36 in Pereira, where motorcycles carry nine tenths. The policy sets the rest, and in the opposite direction to equity. **Greedy** drives 2.94 km per delivered order against 2.25–2.33 for the six order-centric policies, 30% more, because it sends the least-served free courier rather than the nearest one; it shifts distance towards bicycles and walkers (20% of its kilometres, as much as **ILP** and **MIP-Fair** and twice the share of the BRKGA family and **Hierarchical**), so its emissions per kilometre are as low as those of the two ILP-based policies (93–95 g against 104–108 g), but the extra distance dominates: 0.280 kg per delivered order against 0.213–0.216 for **MIP-Fair** and **ILP**, 24% less, and 0.234–0.249 for the BRKGA family and **Hierarchical**. The two ILP-based policies are the cleanest because they drive less and use the slow vehicles more; bundling plays no part, since bundle sizes are 1.00–1.03 for every policy. Equity and emissions therefore pull in opposite directions at this scale: the policy that spreads work also spreads distance. In summary, the regime bounds coverage, the policy sets equity, the threshold does not reorder the policies within its interval, and the fleet composition sets the emissions level while the policy sets the rest.

7. Findings and Discussion

This section addresses the three research questions and discusses their implications.

We begin by addressing the first research question concerning whether an end-of-day guarantee requires the equity history in the state. The results show that fairness enforced within each window does not deliver fairness at the end of the day: the **Hierarchical** policy balances the pay offered among the free couriers of every round and ends the day with a Gini coefficient of 0.53 over the eligible couriers against 0.30 for **Greedy**, because the per-window statistic is computed on the couriers of the round and ignores both what each has already earned and who was left out of the round. Reading $\mathcal{H}_t$ is necessary but not sufficient: the three policies that read it ended the day as unequal as those that do not, because they read a productivity rate, earnings per busy hour, over the couriers of the round rather than exposure over the population of the guarantee. The content of the history must be exposure, earnings per hour connected, and its population the eligible couriers, so that a courier with little work over a long connection is read as under-served rather than as productive; and the idle couriers, whose deficit is one under either reading, must reach the solver through the candidate set, which is the signal that the courier-centric dispatcher obtains for free by serving the least-served courier first. The second generation of X3 confirms the diagnosis: with exposure as the content and the deficit-aware prefilter as the way to reach the idle couriers, **BRKGA-Fair** and **MIP-Fair** close 42–56% of the gap with $k_d = 25$ and **BRKGA-Fair** 56–77% with $k_d = 50$, at no coverage cost.

We next address the second research question concerning which lever satisfies the guarantee at the lowest cost. On the ten seeds of X1 the answer is the assignment rule itself: the courier-centric rule meets $\alpha = 0.40$ on every day under auto-accept and on 98% of the days under the manual regime at a cost, under manual acceptance, of 1.2 points of coverage against the best order-centric policy and, under both regimes, of 0.5–1.8 minutes of lateness, while the order-centric policies meet it on none. Among the three proposed policies, **BRKGA-Fair** is the only one whose parameter moves the end-of-day statistic in the right direction at no coverage cost; the α of **MIP-Fair** is inert because coverage has priority in every round; and **Pay-BRKGA** is the only policy that moves pay, which lowers the platform's bill by 12% (manual) to 20% (auto-accept) of the tariff but does not reach the couriers who are left idle. The second generation of X3 narrows but does not close this difference: reading exposure instead of activity and offering the most-deprived free couriers to the solver halves the gap to **Greedy** ($F_{\text{Gini}} = 0.44$ manual, 0.40–0.42 auto-accept) at no coverage cost and 0.1–0.3 minutes of lateness, so a second-generation **BRKGA-Fair** meets $\alpha = 0.45$ on 62–67% of the days while keeping the point of coverage that **Greedy** gives up. Part of the remaining gap is a matter of candidates and part is structural: with $k_d = 50$ the heuristic reaches 0.40 (manual) and 0.35 (auto-accept), whereas the exact round model flattens, because an order-centric round assigns at most one route per courier and the coverage objective fixes which couriers it needs. Because coverage is flat across policies at the scale of a dispatching zone, the cost of the guarantee is paid

mainly in lateness and in who receives the routes. Dispatching alone cannot lift couriers to the statutory minimum wage at the current tariff; what it can do is distribute the shortfall evenly, which is the practical content of using F_{Gini} as the constraint and F_{Deficit} as a diagnostic.

Finally, we address the third research question concerning the operational conditions under which the guarantee is affordable. The acceptance regime matters more than the policy in the manual world, where coverage is bounded by behaviour; the sanctions that moved the workforce to auto-accept are the platform's way of removing that bound, and our results quantify what the bound costs. Within the confidence interval of the behavioural threshold the fairness ranking of the policies does not change, and the coverage ranking changes only at the upper end, where the courier-centric dispatcher, which does not re-optimise after a rejection, loses most (9% of its coverage against 5.5–7.3% for the policies that do). The three criteria for α give an interval rather than a point; 0.45 is what the order-centric policies currently meet, above the band of criterion B, so the choice between them and the courier-centric rule at 0.35 is a managerial trade-off of a point of coverage and one to two minutes of lateness against ten points of Gini. From a managerial perspective, the central implication is that a fairness guarantee should be stated, measured and enforced on the day, not on the window, and that the pay lever and the assignment lever are complements: the first buys acceptance where couriers reject, the second distributes work where they do not.

8. Conclusion

In this paper, we study a dispatching problem in which a platform repeatedly offers work and pay to self-employed agents who may reject it, and in which the platform is accountable for the distribution of the agents' earnings at the end of the day. We examine this structure in meal delivery, where we formulate the problem as a constrained Markov decision process with an equity history in the state, an end-of-day Gini constraint on hourly earnings over the couriers connected for at least one hour, idle ones included, and acceptance that responds to pay per kilometre in one of two regimes. As a methodological contribution, we develop three cost function approximations that read the equity history in every round, including a linearisation of the Gini constraint that is exact by blocks of courier pairs, and we evaluate them against four benchmarks on 24 real operating days from six Colombian cities with field-calibrated behaviour. Over the eligible couriers the courier-centric dispatcher, which serves the least-served courier first, ends the day at a Gini coefficient of 0.30 and gives work to 97% of them, while every order-centric policy, including the three proposed, ends between 0.52 and 0.57 under manual acceptance (0.50–0.57 under auto-accept) by leaving a quarter of them idle; restricted to the couriers who delivered, the same policies appear more equitable than the corporate one. Reading exposure over the eligible couriers and offering the most-deprived couriers to the solver closes 42–56% of that gap with 25 extra candidates per route and, for **BRKGA-Fair**, 56–77% with 50, at no coverage cost. From a managerial perspective, the

central implication is that a fairness guarantee is a property of the day: it should be measured on hourly earnings over a population the dispatcher cannot shrink, enforced through what every courier has earned per hour connected so far, and complemented by the pay lever where couriers reject.

Beyond meal delivery, the equity history and the end-of-day constraint apply to any sequential assignment problem in which the agents' outcomes accumulate over a horizon and the regulator or the platform cares about their distribution at its end, such as ride-hailing, crowdsourced delivery and task platforms. Within this architecture, the fairness metric, the acceptance model and the per-round problem can each be replaced; the block linearisation applies to any constraint in mean absolute differences in which each agent's outcome takes finitely many values per round, and only the per-window problem must be reformulated for the routing decisions and constraints of the new application.

The study also has several limitations related to the decision environment. The behavioural evidence is from Bogotá, and its use in the other five cities is a transfer assumption; the five cities are robustness checks, not independent calibrations. The acceptance logit rests on 32 decisions by 11 interviewees, with wide intervals, and reflects the 2020 manual regime; we report the sensitivity over the interval of the threshold and run both regimes, but the slope interval was not swept and the parameters are grade C. The 3-km zone keeps the couriers who connect inside it and only the orders dropped inside it, so the simulated fleet is over-supplied by a factor of about three relative to the field (0.36 against 1.2 orders per connected courier-hour); this bounds coverage near 0.79 for every policy, inflates the idle share on which the central result rests and makes the minimum-wage finding partly a property of the instances; a sensitivity with the fleet thinned to the field ratio, and a courier-centric benchmark with arrival order instead of least-served first, are the first items of future work. The 24 instances are 3–5 days from six cities, which the standard errors treat as exchangeable; a city-blocked bootstrap would be more conservative. The survey reports pay in bands, and 93% of couriers report that the pay shown changes after acceptance (realised-to-shown ratio 0.60); the model responds to the pay shown, so realised earnings may be lower. The equity history is intraday whereas the platform's bonus ladder is a multi-day mechanism, and the model has no sanctions, no acceptance-rate dynamics, no cash-handling constraint and no heterogeneity by seniority or nationality. Circuity, vehicle speeds, the 30-minute slack and the UK emission factors are declared assumptions without local sources. Cross-zone effects are not modelled. Future research could extend the history to several days, add the dynamics of sanctions and acceptance rates, and calibrate the behavioural model in a second city.

There are also methodological limitations that motivate further research. **MIP-Fair** uses a candidate prefilter of eight couriers per route for budget reasons and a 10-second limit per round; a column-generation or a rolling subset of pairs could relax both. The three policies are tuned by enumeration over a grid; learning the deficit weight or the pay multiplier as a function of the state, as in Stein et al. (2025), could reduce the cost of the guarantee further, at the price of an

offline training effort. The minimum-wage constraint is not imposed because it is infeasible at the current tariff; a joint design of tariff and dispatching, in which the platform's budget is a decision, is a natural extension. Finally, the 240 public instances of Reyes et al. (2018) could be run with the policies already in the simulator to test the findings outside Colombia.

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Appendix

In this Appendix, we present a more detailed literature review, an illustrative example of the decision process, the complete formulation of the per-round programme of **MIP-Fair**, and additional computational results.

Appendix A. Detailed literature review

This section reviews the literature related to the three components of the problem: the operational MDRP, fairness and compensation in delivery platforms, and courier behaviour. Table 1 in the main body provides a systematic summary of their similarities and differences relative to our work. [TODO: Move the fuller review of the thesis here (operational, economic and behavioural blocks), one subsection each, closing with the gap; switch citations to `citeAppx` (multibib, commented out in the preamble) and add ReferencesAppx.bib so that the appendix references do not count against the page limit, as in the director's manuscripts.]

Appendix A.1. Operational block

Reyes et al. (2018) defined the MDRP as a rolling-horizon, dynamic-deterministic problem and released the Grubhub instance library; Yıldız and Savelsbergh (2019) obtained provably high-quality solutions for a deterministic variant; Steever et al. (2019) revise assignments on every event; Ulmer et al. (2021) anticipate random ready times with approximate dynamic programming; and later contributions address shared logistics (Bi et al. 2023), courier operating regions (Auad et al. 2023), batching for crowdsourced fleets (Simoni and Winkenbach 2023), ghost kitchens (Neria et al. 2025) and on-demand delivery from stores (Liu and Luo 2023). In all of these, acceptance is perfect or exogenous. [TODO: expand.]

Appendix A.2. Economic block

Most operational models pay a fixed amount per delivery or minimise distance as a cost proxy. Reyes et al. (2018) and Yıldız and Savelsbergh (2019) include a shift-level guaranteed minimum set by the company; Wen et al. (2010) balance workload and income opportunities within a multi-period routing model; Lesmana et al. (2019) trace the efficiency–fairness frontier of ride-sourcing; Bajec and Tuljak-Suban (2022) use Shapley-value incentives; Sobhanan et al. (2024) give equity–cost frontiers; Gupta et al. (2022), Nair et al. (2022) and Martínez-Sykora et al. (2024) bring fairness to food delivery. None integrates acceptance-dependent pay, a real tariff and an intertemporal guarantee within a rolling-horizon routing problem. [TODO: expand.]

Appendix A.3. Behavioural block

Liao et al. (2020) balance workload and argue that equitable distribution is an incentive; reinforcement-learning approaches (Ahamed et al. 2021, Bozanta et al. 2022) include availability in the state but keep acceptance exogenous; Haferkamp et al. (2024) address repositioning; Quintero Rojas (2020) makes acceptance depend on distance. The fair-machine-learning literature shows that short- and long-horizon fairness diverge (Corbett-Davies et al. 2023). [TODO: expand; add TODO authors (2025).]

Appendix B. Illustrative example and algorithmic details

In this section, we first provide an illustrative numerical example connecting the sequential decision process to the dispatching decisions faced by the platform. We then present the complete per-round formulation of **MIP-Fair** and the tuning of the policy parameters.

Appendix B.1. Notation

Table B.4 collects the symbols used in Sections 3 and 4.

Symbol	Meaning
$t \in T = \{0, \dots, t_{\max}\}$	decision points (one per two-minute window); $t_{\max}$ the last one
$S_t = (\mathcal{L}_t, \mathcal{O}_t, \mathcal{P}_t, \mathcal{H}_t)$	state: couriers, unassigned orders, preparation estimates, equity history
$h_c(t) = (\text{earn}_c, \text{ord}_c, \text{busy}_c)$	earnings, orders delivered and busy minutes of courier c up to t
$\text{conn}_c(t)$, $w_c(t)$	connected time since log-in; hourly earnings so far, $\text{earn}_c(t)/\text{conn}_c(t)$
C, C^e, C_t^{free}, C_t	couriers; eligible (connected ≥ 1 h); available at t; connected at t
x_t, $\mathcal{S}_t$, K_t	decision (courier, route, pay triples); feasible routes; candidate pairs
$\rho_{c,s}$, ρ_0, ρ_1, m_c, δ	pay offered; base fee and per-km rate of the tariff; pay multiplier and its band
$z_{c,s}$, $\mathbf{z}_t$, θ, b	acceptance indicator and vector; logit threshold (COP/km) and slope
ω_{t+1}, $\mathcal{T}$, $\mathcal{C}$, c_1, c_2, λ	exogenous information; transition; stage cost and its weights; payment weight
F_{Gini}, $F_{\text{Gini}}^{\text{act}}$, F_{MaxMin}, F_{Orders}, F_{Deficit}, α, w^*	Gini over eligible / active couriers, max–min, order-count Gini, minimum-wage deficit, tolerated inequality, minimum wage
k, k_d, K	nearest-courier prefilter; deficit-aware prefilter; couriers evaluated by the decoder per route
γ, λ_{pay}, λ_{def}, λ_{tt}, μ	deficit weight of **BRKGA-Fair**; pay and deficit weights of **Pay-BRKGA**; travel-time and lateness weights of **MIP-Fair**
v, W_{ref}, d_{ij}	Gini excess, its scale and the pairwise differences of **MIP-Fair**
κ, $\nu(v)$, Q, $\Delta^{\max}$	circuity factor, vehicle speed, route capacity, slack on the promised time

Table B.4: Notation

Appendix B.2. Illustrative numerical example

We use a two-courier example to connect the components of Section 3.2 with the dispatching decisions and to show why a policy that is fair within every window can be unfair at the end of the day. Couriers A and B are connected from 12:00 to 13:00, so $\text{act}_A(T) = \text{act}_B(T) = 60$ minutes whatever is dispatched; the tariff is flat at 2,200 COP per order; B is slightly farther from the store than A. Four routes become available during the hour: at 12:00 a bundle of three orders from one store (pay 6,600 COP, duration 30 minutes including service), and at 12:30, 12:40 and 12:50 one singleton each (pay 2,200 COP, duration 8 minutes, so that the courier who takes one is free again before the next window). Coverage is the first priority of every policy, so all four routes are dispatched in every case; the question is to whom.

At $t =$ 12:00 the state is S_0 with both couriers idle and $h_A(0) = h_B(0) = (0, 0, 0)$. Both the per-window policy and the history-aware policy dispatch the bundle to A, the nearer courier, since

nothing distinguishes the two couriers yet. At $t = 12{:}30$ the state contains $h_A = (6{,}600, 3, 30)$ and $h_B = (0, 0, 30)$, that is, hourly earnings so far of $w_A(t) = 13{,}200$ COP/h and $w_B(t) = 0$, and one singleton is available. Consider two decisions. The per-window policy balances the pay offered within the window among the free couriers: with a single route, the window statistic is the same whichever courier receives it (one courier is offered 2,200 and the other 0), so the tie is broken by distance and the singleton goes to A. The same happens at 12:40 and 12:50. The end-of-day hourly earnings are $w_A = 6{,}600 + 3 \times 2{,}200 = 13{,}200$ COP/h and $w_B = 0$, and by Eq. (7)

$$F_{\text{Gini}} = \frac{2 \times |13{,}200 - 0|}{2 \times 2 \times 13{,}200} = 0.5.$$

The history-aware policy reads $\mathcal{H}_t$ at 12:30: $\text{deficit}_B = 1$ and $\text{deficit}_A = 0$ in Eq. (11), or equivalently $w_B(s) = 3{,}474$ and $w_A(s) = 13{,}895$ COP/h in Eq. (12) (projected to the end of the window), so the singleton goes to B, and likewise at 12:40 and 12:50. The end-of-day hourly earnings are $w_A = w_B = 6{,}600$ COP/h and $F_{\text{Gini}} = 0$, with the same coverage (six orders), the same active time, and a difference in travel of a few hundred metres.

This example highlights two points. First, the per-window statistic cannot distinguish the two decisions: within each window after 12:00 the pay offered is $(2{,}200, 0)$ whichever courier is chosen, so a policy that is fair within the window, such as the **Hierarchical** benchmark, is indifferent between an end-of-day Gini coefficient of 0.5 and one of 0. Second, what distinguishes them is the component $\mathcal{H}_t$ of the state, which records that A has already earned 6,600 COP during the same active time in which B has earned nothing. The effect compounds when couriers connect at different times, when routes dispatched in different windows differ in value, and when some couriers are systematically nearer to the stores, which is why the **Hierarchical** policy raises the end-of-day Gini coefficient on real days (Section 6.1).

Appendix B.3. The BRKGA decoder

Algorithm 2 is the decoder shared by **BRKGA**, **BRKGA-Fair** and **Pay-BRKGA** (Section 4); the policies differ only in the score it minimises.

Appendix B.4. Per-round formulation of MIP-Fair

At round t, let C_t be the set of connected couriers, $C_t^{\text{free}} \subseteq C_t$ the free ones, K_t the candidate pairs and $\mathcal{O}_t^{\text{ready}}$ the ready orders. Let $x_{cs} \in \{0, 1\}$ for $(c, s) \in K_t$, $y_o \in \{0, 1\}$ for $o \in \mathcal{O}_t^{\text{ready}}$, $w_c \geq 0$ for $c \in C_t$, $d_{ij} \geq 0$ for free–free pairs $i < j$, $\text{late}_o \geq 0$ and $v \geq 0$. The three stages of Eq. (14) are solved over the following constraints:

Algorithm 2 BRKGA decoder shared by **BRKGA**, **BRKGA-Fair** and **Pay-BRKGA**

Input: keys κ_o for ready orders, κ_c for available couriers, candidate pairs K_t, score function $\text{score}(s, c)$

Output: assignment A

$A \leftarrow \emptyset$; mark all couriers free
for each store r in decreasing order of $\max_{o: r_o = r} \kappa_o$ **do**
  for each order o of r in decreasing order of κ_o, not yet assigned **do**
    $\mathcal{S}_o \leftarrow$ candidate routes headed by o (singleton and the store bundle containing o)
    $\mathcal{C}_o \leftarrow$ the K free couriers with highest κ_c among $\{c : (c, s) \in K_t,\ s \in \mathcal{S}_o\}$
    if $\mathcal{C}_o \neq \emptyset$ **then**
      $(s^*, c^*) \leftarrow \arg\min\{\text{score}(s, c) : s \in \mathcal{S}_o,\ c \in \mathcal{C}_o,\ (c, s) \in K_t\}$
      $A \leftarrow A \cup \{(c^*, s^*)\}$; mark c^* busy and the orders of s^* assigned
return A

$$\sum_{(c,s) \in K_t : o \in s} x_{cs} \geq y_o, \qquad \forall o \tag{B.1}$$

$$\sum_{(c,s) \in K_t : o \in s} x_{cs} \leq 1, \qquad \forall o \tag{B.2}$$

$$\sum_{s : (c,s) \in K_t} x_{cs} \leq 1, \qquad \forall c \tag{B.3}$$

$$w_c = w_c^0 \Big(1 - \sum_s x_{cs}\Big) + \sum_s x_{cs}\, w_c(s), \qquad \forall c \tag{B.4}$$

$$w_c = w_c^0, \qquad \forall c \tag{B.5}$$

$$d_{ij} \geq w_i - w_j, \quad d_{ij} \geq w_j - w_i, \qquad \forall i \tag{B.6}$$

$$\sum_{i < j \in C_t^{\text{free}}} d_{ij} + \sum_{i \in C_t^{\text{free}},\, j \notin C_t^{\text{free}}} \Big[(1 - \textstyle\sum_s x_{is}) |w_i^0 - w_j^0| + \sum_s x_{is} |w_i(s) - w_j^0| \Big] + \sum_{i < j,\ i, j \notin C_t^{\text{free}}} |w_i^0 - w_j^0| - \alpha\, n \sum_{c \in C_t} w_c \leq n\, W_{\text{ref}}\, v \tag{B.7}$$

$$\text{late}_o \geq \sum_{(c,s) \in K_t : o \in s} x_{cs}\ \max(0, \hat{\tau}_{\text{drop}}(c, s, o) - d_o), \qquad \forall o \tag{B.8}$$

$$x_{cs} \in \{0, 1\},\ y_o \in \{0, 1\},\ d_{ij}, w_c, \text{late}_o, v \geq 0. \tag{B.9}$$

Eq. (B.1) links coverage to dispatched routes and Eqs. (B.2) - (B.3) ensure that each order is in at most one dispatched route and each courier receives at most one. Eqs. (B.4) - (B.5) give the projected hourly earnings, linear in x for free couriers and fixed for the others. Eq. (B.6) defines the pairwise differences for free–free pairs only, and Eq. (B.7) is the Gini constraint of Eq. (13) with the free–fixed block written out as linear terms and the fixed–fixed block as a constant; $n = |C_t|$ and W_{ref} is the mean of w_c^0 over C_t (or w^* when that mean is zero). Eq. (B.8) computes the projected lateness of each order under its dispatched route, and Eq. (B.9) defines the domains. Stage 1 maximises $\sum_o y_o$; stage 2 fixes that value and minimises v; stage 3 fixes v^* and minimises $\lambda_{\text{tt}} \sum_{(c,s)} \tau_s x_{cs} + \mu \sum_o \text{late}_o$. We use $\lambda_{\text{tt}} = 0.01$ per minute of travel and $\mu = 0.02$ per minute of lateness, the tie-breaking weights of the **ILP** and **Hierarchical** policies.

Appendix B.5. Tuning of the policy parameters

Table C.14 in Appendix C.6 reports the two tuning ablations on the 24 instances: the BRKGA with population 20 and 20 generations leaves coverage unchanged and raises the end-of-day Gini coefficient by 0.007 at almost three times the CPU time, and the **MIP-Fair** prefilter with $k = 25$ lowers it by 0.039 and raises coverage by 0.007 at eight times the CPU time (31 minutes per day on average, 227 at most, and beyond the four-hour timeout on the largest Bogotá day on four or five of the seeds, depending on the configuration); the deficit-aware prefilter of Section 6.1 obtains more than twice that reduction for the same order of CPU time. The settings of Section 5.5 are therefore kept. In the pilot, the BRKGA-type policies take 9–10 seconds for a full day of Pereira and 30–35 seconds for a two-hour window of Bogotá, the **Hierarchical** policy 16 and 39 seconds, and **MIP-Fair** with $k = 8$ 36 seconds and 4.5 minutes, respectively; a BRKGA with population 20 and 20 generations did not change coverage on Pereira.

Appendix B.6. Verification against enumeration

The four pieces (BRKGA port, **BRKGA-Fair**, **MIP-Fair**, **Pay-BRKGA**) are covered by unit tests against exhaustive enumeration on a toy instance of three couriers and four orders: the candidate sets coincide with those of the **ILP** and the **Hierarchical** policy (18 pairs); the $\text{score}_{\text{fpo}}$ decoder reaches the exhaustive optimum of its fitness; **MIP-Fair** with $\alpha = 0$ coincides with the exhaustive lexicographic optimum under both priorities and both earnings bases, and with $\alpha = 1$ covers 4 of 4 orders like the **ILP**; the block linearisation reproduces the all-pairs model; and for **Pay-BRKGA**, with λ_{pay} alone all offers fall below the tariff, with λ_{def} alone the under-paid courier receives an offer above the tariff and leaves the deficit, and expected coverage equals coverage times the acceptance rate under a fixed-probability model.

Appendix C. Additional results of the experiments

In this section, we provide the detail that complements the main results: the instances, parameters and factors of the study, the results by city, the second generation paired with its parents,

the metrics not reported in the main tables, the compliance and the earnings by population, the ablations of X3, the Gini coefficient under the sensitivity settings together with the regime loss and the choice of α, and the emissions diagnostic.

Appendix C.1. Instances, parameters and factors

Table C.6 lists the 24 instances described in Section 5.1, Table C.5 the parameters of the simulation with their values, sources and evidence grade, and Table C.7 the factors and levels of the computational study of Section 5.3; Figure C.3 shows a 3-km zone and Figure C.4 the acceptance curve of Eq. (4).

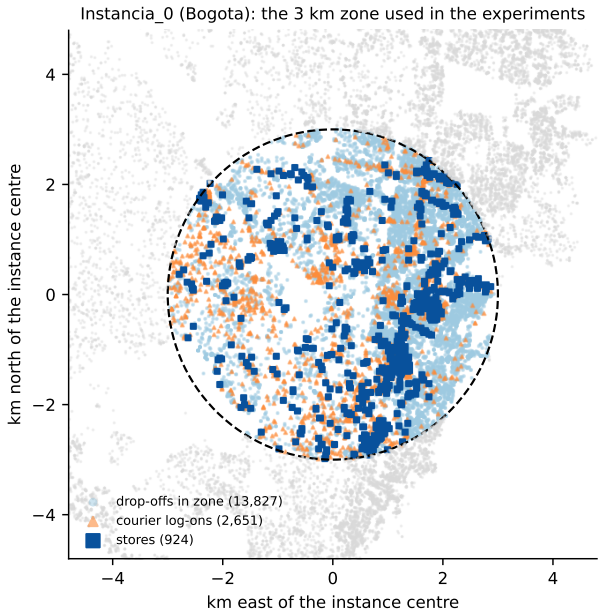


Figure C.3: A 3-km dispatching zone in Bogotá

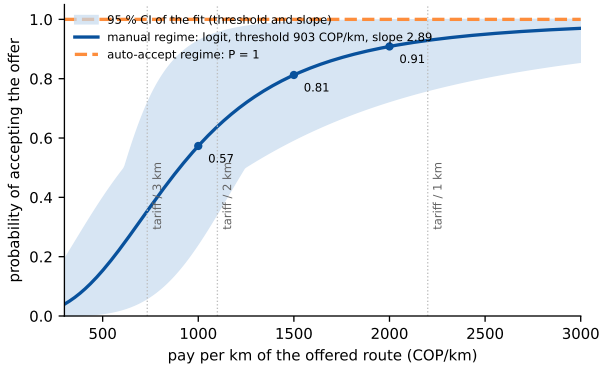


Figure C.4: Acceptance probability as a function of pay per kilometre

Parameter	Value	Source	Grade
Acceptance threshold θ	903 COP/km (95% [615; 1,242])	interviews, 32 decisions / 11 couriers	C
Acceptance slope b	2.89 (95% [2.02; 5.23])	interviews	C
Acceptance regime	manual / auto-accept	interviews (A), survey 64%	A
Tariff ρ_0	2,200 COP per order, flat	survey ($n = 132$)	B
Minimum wage w^*	4,367.91 COP/h	statutory 2021 / (48 h × 4.33)	–
In-store service	20 min (base), 60 min (stress)	survey / interviews	B / C
Drop-off service	4 min	public instances	–
Decision delay	16 ± 8 s	interviews (IQR 9–28 s)	C
Speeds (bike/moto/walk/car)	15 / 25 / 5 / 20 km/h	assumption	–
Circuity κ	1.3 (1.0 sensitivity)	assumption	–
Hard deadline $\Delta^{\max}$	promised time + 30 min	assumption (interviews)	–
Route capacity Q	3	simulator default	C
Window	2 min	platform practice	–
Emission factors	moto 0.114, car 0.167, bike/walk 0 kg CO_2e/km	Department for Energy Security and Net Zero and Department for Environment, Food & Rural Affairs (UK) (2023)	–

Table C.5: Parameters, values, sources and evidence grade (A: consistent across at least 15 interviewees or clear convergence of both sources; B: 5–14 interviewees or a single solid source; C: fewer than 5, or a small-n estimate with a wide interval)

Inst.	City	Orders	Couriers	Stores	Orders/courier	Vehicle mix (full city, %: bike/moto/walk/car)
0	Bogotá	10,215	2,650	926	3.9	45/39/9/7
6	Bogotá	9,743	2,489	997	3.9	43/39/10/8
12	Bogotá	5,618	1,967	859	2.9	44/41/6/9
18	Bogotá	5,875	1,931	912	3.0	39/39/11/11
2	Medellín	6,169	1,692	602	3.6	16/72/6/7
8	Medellín	5,778	1,773	628	3.3	15/71/7/7
14	Medellín	3,953	1,244	614	3.2	15/72/5/8
20	Medellín	3,918	1,259	587	3.1	15/70/7/8
5	Barranquilla	7,773	1,442	470	5.4	21/64/4/10
11	Barranquilla	7,116	1,532	474	4.6	22/63/4/11
17	Barranquilla	3,606	1,168	4,014[a]	3.1	19/66/3/13
23	Barranquilla	3,612	1,046	431	3.5	20/59/5/15
7	Cali	1,813	745	269	2.4	21/72/2/5
13	Cali	1,262	488	260	2.6	21/73/2/5
19	Cali	1,292	476	264	2.7	22/71/2/5
1	Bucaramanga	1,851	574	216	3.2	9/89/1/1
3	Bucaramanga	1,851	574	216	3.2	9/89/1/1
9	Bucaramanga	1,527	610	218	2.5	9/89/1/1
15	Bucaramanga	976	536	202	1.8	7/90/1/1
21	Bucaramanga	948	498	204	1.9	9/87/2/1
4	Pereira	568	253	135	2.2	5/91/2/2
10	Pereira	498	277	140	1.8	6/90/2/2
16	Pereira	342	268	401[a]	1.3	5/90/2/3
22	Pereira	374	246	122	1.5	6/89/3/3

[a] One store record per order in the export; does not affect the simulation. Instances 1 and 3 are different days with identical counts.

Table C.6: The 24 instances: orders, couriers and stores in the 3-km zone, and vehicle mix of the full city

Factor	Levels	Acts on
Policy	**Greedy**, **ILP**, **BRKGA**, **Hierarchical**; **BRKGA-Fair** $w \in \{5, 10, 20, 40\}$ min; **MIP-Fair** $\alpha \in \{0.20, 0.25, 0.30, 0.35, 0.40, 1.00\}$; **Pay-BRKGA** $\lambda_{\text{pay}} \in \{0.1, 0.2, 0.5, 1.0\}$	round
Acceptance regime	manual, auto-accept	Eq. (4)
Logit threshold θ (X2, manual regime)	615 / 903 / 1,242 COP/km, slope fixed at 2.89	Eq. (4)
Earnings basis and deficit prefilter k_d (X3)	activity / exposure; $k_d \in \{0, 8, 25, 50\}$	round
Instance	24 (six cities), 3-km zone	block
Seed	10 in X1, five in X2 and X3, the same in every cell	acceptance, delays, BRKGA

Table C.7: Factors and levels of the computational study

Appendix C.2. Results by city

Table C.8 repeats the manual-regime block of Table 2 for each of the six cities, with the compliance with $\alpha = 0.40$ as the last column. The central result holds city by city: **Greedy** is between 0.275 (Medellín) and 0.323 (Pereira) on the end-of-day Gini coefficient over the eligible couriers and meets $\alpha = 0.40$ on 85–100% of the days in every city (Pereira lowest), while the six order-centric policies are between 0.448 (**ILP**, Bogotá and Medellín) and 0.637 (**BRKGA**, Bucaramanga and Pereira) and meet it on none. The guarantee is therefore uniformly within reach of the courier-centric rule, and the question of where it is cheap or expensive becomes the question of where the order-centric policies fall furthest from it. They fall furthest in the two smallest fleets, Bucaramanga and Pereira (0.56–0.64, with only 59–73% of the eligible couriers active), and closest in Bogotá and Medellín (0.45–0.51, 79–86% active), where the fleet is large relative to demand and the best-placed courier changes more often. The coverage that **Greedy** gives up with respect to the best policy of the city is between 0.8 points (Bogotá) and 1.5 points (Cali), and its extra lateness against **Hierarchical**, the least late policy in every city, between 1.3 (Bucaramanga) and 2.3 minutes (Bogotá); the pay lever of **Pay-BRKGA** cuts the bill most where the fleet is small and routes are long (Pereira, 19%; Bucaramanga, 15%) and least in Bogotá and Medellín (7–9%). The second-generation variants with $k_d = 25$ (five seeds) keep the ordering: under the manual regime they reach 0.37–0.38 in Barranquilla, Bogotá and Medellín, 0.45–0.47 in Cali and 0.49–0.55 in Bucaramanga and Pereira, so the gap to **Greedy** that remains is again largest in the two smallest fleets.

Appendix C.3. The second generation against its parents

Table C.9 reports the means of the second-generation configurations of X3 against their first-generation parents and **Greedy** on the same five seeds, and Table C.10 reports, for every configuration, the paired difference with its first-generation parent on the same (instance, seed) pairs, the share of the gap to **Greedy** closed (ratio of means, Section 5.4) and the compliance with $\alpha = 0.40$ and 0.45. The exposure basis alone moves F_{Gini} by 0.004–0.025; the prefilter moves it by 0.054–0.133, with every interval excluding zero and coverage within 0.006 of the parent.

Appendix C.4. Lateness frontier and metrics not reported in the main tables

Figure C.5 is the lateness counterpart of Figure 1: the three sweeps occupy the same narrow band of the Gini axis, **Hierarchical** is the least late policy and **Greedy** the latest, and the second-generation variants stay within 0.3 minutes of their parents.

Table C.11 reports, for the configurations of Table 2, the max–min fairness F_{MaxMin} (the lowest hourly earnings among the active couriers), the click-to-door time of the delivered orders (mean and 90th percentile) and the solve time per round (mean, and the maximum over all rounds and runs). Under the flat tariff the order-count Gini coefficient F_{Orders} coincides with $F^{\text{act}}_{\text{Gini}}$ for every policy that pays the tariff and is therefore not repeated. Click-to-door differs by at most two

Policy	Coverage	End-of-day Gini F_{Gini}	Active eligible	Lateness (min)	Payment (% of tariff)	$F_{\text{Gini}} \leq 0.40$ (% of days)
Barranquilla (4 instances)						
Greedy	0.692	**0.304**	**0.98**	20.0	100	100
ILP	0.670	0.470	0.80	19.8	100	0
Hierarchical	0.700	0.491	0.80	**18.2**	100	0
BRKGA	0.701	0.513	0.76	18.7	100	0
BRKGA-Fair ($\gamma = 40$)	0.701	0.478	0.82	18.9	100	0
MIP-Fair ($\alpha = 0.30$)	0.690	0.503	0.78	19.5	100	0
Pay-BRKGA ($\lambda_{\text{pay}} = 0.5$)	0.690	0.499	0.78	19.0	92	0
Bogotá (4 instances)						
Greedy	0.811	**0.282**	**0.96**	20.1	100	100
ILP	0.799	0.448	0.85	19.6	100	0
Hierarchical	0.815	0.503	0.81	**17.8**	100	0
BRKGA	0.817	0.509	0.79	18.5	100	0
BRKGA-Fair ($\gamma = 40$)	**0.819**	0.467	0.87	18.8	100	0
MIP-Fair ($\alpha = 0.30$)	0.808	0.484	0.82	19.4	100	0
Pay-BRKGA ($\lambda_{\text{pay}} = 0.5$)	0.809	0.493	0.81	18.8	93	0
Bucaramanga (5 instances)						
Greedy	0.730	**0.306**	**0.97**	19.6	100	100
ILP	0.714	0.593	0.63	19.1	100	0
Hierarchical	0.743	0.579	0.71	**18.3**	100	0
BRKGA	0.743	0.637	0.59	18.6	100	0
BRKGA-Fair ($\gamma = 40$)	**0.744**	0.608	0.65	18.6	100	0
MIP-Fair ($\alpha = 0.30$)	0.741	0.625	0.60	18.7	100	0
Pay-BRKGA ($\lambda_{\text{pay}} = 0.5$)	0.726	0.613	0.61	18.8	85	0
Cali (3 instances)						
Greedy	0.748	**0.322**	**0.96**	20.7	100	100
ILP	0.727	0.530	0.70	19.9	100	0
Hierarchical	0.762	0.563	0.71	**18.5**	100	0
BRKGA	0.762	0.591	0.63	18.8	100	0
BRKGA-Fair ($\gamma = 40$)	**0.763**	0.561	0.69	18.9	100	0
MIP-Fair ($\alpha = 0.30$)	0.759	0.554	0.68	19.2	100	0
Pay-BRKGA ($\lambda_{\text{pay}} = 0.5$)	0.743	0.564	0.66	19.2	86	0
Medellín (4 instances)						
Greedy	0.812	**0.275**	**0.99**	18.7	100	100
ILP	0.797	0.448	0.82	18.0	100	0
Hierarchical	0.823	0.457	0.85	**17.0**	100	0
BRKGA	0.824	0.490	0.79	17.4	100	0
BRKGA-Fair ($\gamma = 40$)	0.824	0.461	0.85	17.5	100	0
MIP-Fair ($\alpha = 0.30$)	0.820	0.474	0.82	17.7	100	0
Pay-BRKGA ($\lambda_{\text{pay}} = 0.5$)	0.815	0.475	0.81	17.6	91	0
Pereira (4 instances)						
Greedy	0.762	**0.323**	**0.95**	20.3	100	85
ILP	0.723	0.619	0.59	20.3	100	0
Hierarchical	**0.776**	0.564	0.73	**18.8**	100	0
BRKGA	0.775	0.637	0.61	19.1	100	0
BRKGA-Fair ($\gamma = 40$)	0.775	0.619	0.65	19.1	100	0
MIP-Fair ($\alpha = 0.30$)	0.770	0.610	0.63	19.4	100	0
Pay-BRKGA ($\lambda_{\text{pay}} = 0.5$)	0.742	0.621	0.61	19.4	81	0

Table C.8: Results by city, manual regime: mean over the instances of the city (ten seeds averaged within each instance) for the configurations of Table 2; the last column is the share of (instance, seed) pairs with $F_{\text{Gini}} \leq 0.40$. Best value per column and city in bold

Policy	Coverage mean	Coverage SE	End-of-day Gini F_{Gini} mean	End-of-day Gini F_{Gini} SE	Active eligible	Gini $F^{\text{act}}_{\text{Gini}}$	Lateness (min)	Min.-wage deficit F_{Deficit}	Payment (% of tariff)	CPU time (min/run)
Manual regime										
Greedy	0.758	0.013	**0.301**	0.008	**0.97**	0.507	19.9	0.981	100	**2.2**
BRKGA-Fair ($\gamma = 40$)	0.770	0.013	0.534	0.015	0.75	0.504	18.6	0.986	100	2.4
BRKGA-Fair ($\gamma = 20$, exposure, $k_d = 8$)	0.770	0.013	0.483	0.016	0.80	0.432	18.7	0.993	100	4.6
BRKGA-Fair ($\gamma = 40$, exposure, $k_d = 25$)	0.769	0.013	0.437	0.016	0.85	**0.398**	18.8	0.995	100	4.9
MIP-Fair ($\alpha = 0.30$)	0.763	0.013	0.544	0.013	0.72	0.458	19.0	0.988	100	4.6
MIP-Fair ($\alpha = 0.30$, exposure, $k_d = 8$, $\varepsilon = 0.00$)	0.769	0.013	0.465	0.012	0.86	0.473	19.1	0.990	100	11.3
MIP-Fair ($\alpha = 0.30$, exposure, $k_d = 25$, $\varepsilon = 0.00$)	0.767	0.013	0.435	0.014	0.88	0.466	19.3	0.990	100	23.5
Pay-BRKGA ($\lambda_{\text{pay}} = 0.5$)	0.753	0.013	0.547	0.013	0.71	0.456	18.8	0.991	88	2.8
Pay-BRKGA ($\lambda_{\text{pay}} = 0.5$, exposure, $k_d = 8$)	0.753	0.013	0.526	0.014	0.74	0.445	18.9	0.992	88	5.2
Auto-accept regime										
Greedy	0.786	0.013	**0.296**	0.008	**0.97**	0.586	19.0	0.973	100	**1.8**
BRKGA-Fair ($\gamma = 40$)	0.786	0.013	0.536	0.015	0.76	0.512	18.2	0.983	100	2.0
BRKGA-Fair ($\gamma = 20$, exposure, $k_d = 8$)	0.786	0.013	0.482	0.016	0.80	0.442	18.2	0.992	100	4.1
BRKGA-Fair ($\gamma = 40$, exposure, $k_d = 25$)	0.786	0.013	0.416	0.016	0.87	**0.390**	18.3	0.995	100	4.3
MIP-Fair ($\alpha = 0.30$)	0.786	0.013	0.534	0.015	0.73	0.448	18.8	0.988	100	3.2
MIP-Fair ($\alpha = 0.30$, exposure, $k_d = 8$, $\varepsilon = 0.00$)	0.786	0.013	0.426	0.011	0.90	0.458	18.8	0.991	100	6.9
MIP-Fair ($\alpha = 0.30$, exposure, $k_d = 25$, $\varepsilon = 0.00$)	0.786	0.013	0.401	0.011	0.91	0.456	18.8	0.990	100	11.1
Pay-BRKGA ($\lambda_{\text{pay}} = 0.5$)	0.786	0.013	0.534	0.013	0.73	0.454	18.3	0.992	80	2.2
Pay-BRKGA ($\lambda_{\text{pay}} = 0.5$, exposure, $k_d = 8$)	0.786	0.013	0.507	0.013	0.77	0.440	18.3	0.994	80	4.2

Table C.9: Second-generation variants (exposure basis over the eligible couriers; deficit-aware prefilter with k_d extra candidates; coverage slack ε in **MIP-Fair**) against their first-generation parents and **Greedy**, on the five seeds of X3 (five of its eight configurations; 1,920 runs in all) and the same seeds of X1: mean over the 24 instances and standard error, columns and bold convention as in Table 2; the $k_d = 8$ row of **BRKGA-Fair** descends from $\gamma = 20$ and the $k_d = 25$ row from $\gamma = 40$

Configuration	n	ΔF_{Gini} vs parent [95% CI]	Δ coverage vs parent [95% CI]	Δ lateness (min)	Gap closed (%)	Compliance (%) $\alpha = 0.40$	Compliance (%) 0.45
Manual regime							
BRKGA-Fair ($\gamma = 20$) exposure basis	120	−0.023 [-0.025, -0.021]	+0.001 [-0.000, +0.001]	+0.0	10	0	28
BRKGA-Fair ($\gamma = 20$) exposure, $k_d = 8$	120	−0.054 [-0.057, -0.051]	−0.000 [-0.001, +0.001]	+0.1	23	13	50
BRKGA-Fair ($\gamma = 40$) exposure, $k_d = 25$	120	−0.097 [-0.101, -0.093]	−0.001 [-0.002, -0.000]	+0.2	42	49	62
MIP-Fair ($\alpha = 0.30$) exposure basis	120	−0.004 [-0.006, -0.003]	+0.000 [-0.001, +0.001]	−0.0	2	0	4
MIP-Fair ($\alpha = 0.30$) exposure, $k_d = 8$	120	−0.079 [-0.083, -0.075]	+0.005 [+0.004, +0.006]	+0.1	33	8	50
MIP-Fair ($\alpha = 0.30$) exposure, $k_d = 8$, $\varepsilon = 0.05$	120	−0.080 [-0.084, -0.076]	+0.006 [+0.005, +0.007]	+0.1	33	8	50
MIP-Fair ($\alpha = 0.30$) exposure, $k_d = 25$	120	−0.109 [-0.114, -0.105]	+0.003 [+0.002, +0.004]	+0.3	45	40	61
Pay-BRKGA ($\lambda_{\text{pay}} = 0.5$) exposure, $k_d = 8$	120	−0.021 [-0.023, -0.019]	+0.000 [-0.001, +0.001]	+0.1	8	0	9
Auto-accept regime							
BRKGA-Fair ($\gamma = 20$) exposure basis	120	−0.025 [-0.027, -0.023]	+0.000 [-0.000, +0.000]	+0.0	10	1	31
BRKGA-Fair ($\gamma = 20$) exposure, $k_d = 8$	120	−0.057 [-0.059, -0.054]	+0.000 [-0.000, +0.000]	+0.0	23	12	50
BRKGA-Fair ($\gamma = 40$) exposure, $k_d = 25$	120	−0.120 [-0.124, -0.115]	+0.000 [-0.000, +0.000]	+0.1	50	56	67
MIP-Fair ($\alpha = 0.30$) exposure basis	120	−0.009 [-0.011, -0.007]	+0.000 [+0.000, +0.000]	−0.0	4	0	8
MIP-Fair ($\alpha = 0.30$) exposure, $k_d = 8$	120	−0.108 [-0.115, -0.101]	+0.000 [+0.000, +0.000]	−0.0	45	35	72
MIP-Fair ($\alpha = 0.30$) exposure, $k_d = 8$, $\varepsilon = 0.05$	120	−0.108 [-0.115, -0.101]	+0.000 [+0.000, +0.000]	−0.0	45	40	72
MIP-Fair ($\alpha = 0.30$) exposure, $k_d = 25$	120	−0.133 [-0.139, -0.127]	+0.000 [+0.000, +0.000]	+0.0	56	63	83
Pay-BRKGA ($\lambda_{\text{pay}} = 0.5$) exposure, $k_d = 8$	120	−0.027 [-0.029, -0.025]	−0.000 [-0.000, +0.000]	+0.0	11	0	14

Table C.10: Second-generation configurations of X3 against their first-generation parents, paired by (instance, seed) on five seeds: paired differences of the end-of-day Gini coefficient, coverage and mean lateness with 95% intervals (± 1.96 SE over the pairs), share of the gap to **Greedy** closed as a ratio of means, and compliance with $\alpha = 0.40$ and 0.45

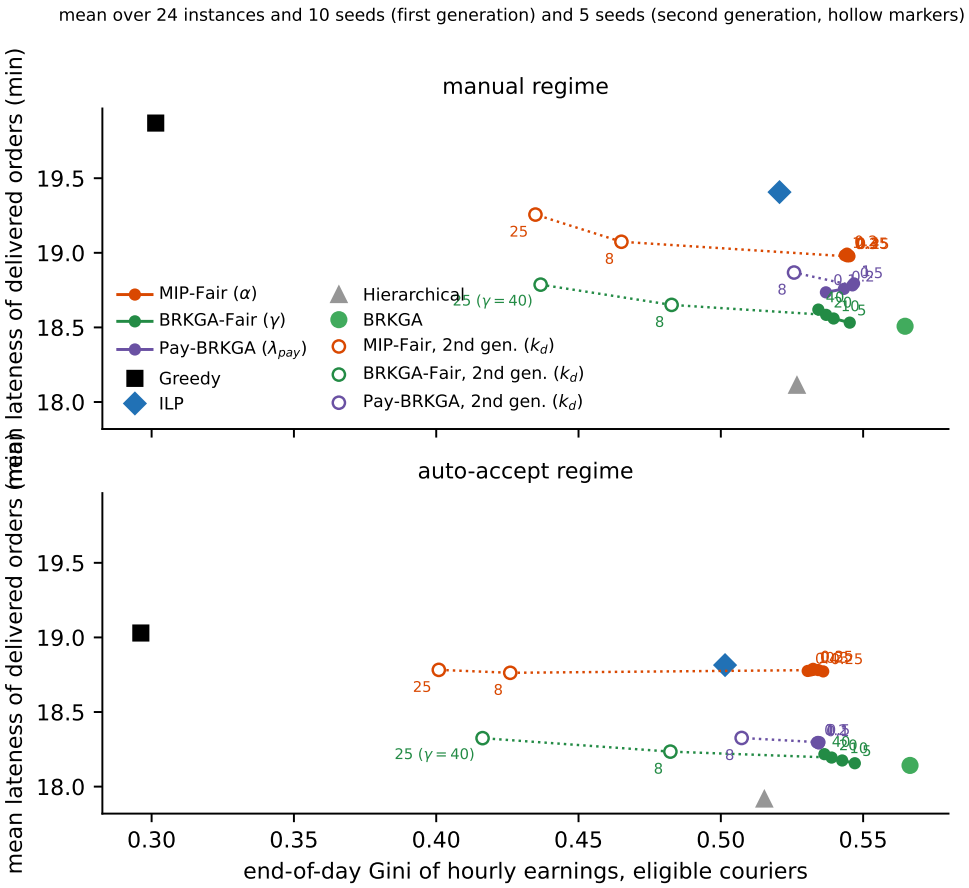


Figure C.5: Mean lateness against the end-of-day Gini coefficient by policy, manual (top) and auto-accept (bottom) regime; conventions as in Figure 1

minutes between policies; every policy solves a round in less than half a second on average and in at most 7.6 s, within the 10-second limit per stage of **MIP-Fair**.

Policy	F_{MaxMin} **(COP/h)**	**Click-to-door (min)**		**Solve time per round (s)**	
		mean	**p90**	**mean**	**max**
Manual regime					
Greedy	195	50.4	59.1	0.18	4.5
ILP	96	49.9	59.0	0.16	3.1
Hierarchical	95	48.7	58.2	0.27	5.6
BRKGA	95	49.1	58.4	0.20	3.3
BRKGA-Fair ($\gamma = 40$)	94	49.2	58.5	0.20	3.5
MIP-Fair ($\alpha = 0.30$)	95	49.5	58.7	0.38	7.6
Pay-BRKGA ($\lambda_{\text{pay}} = 0.5$)	73	49.3	58.6	0.24	3.5
Auto-accept regime					
Greedy	200	49.6	58.7	0.16	3.5
ILP	97	49.4	58.6	0.13	2.7
Hierarchical	97	48.5	58.1	0.25	4.7
BRKGA	95	48.7	58.2	0.18	3.3
BRKGA-Fair ($\gamma = 40$)	94	48.8	58.2	0.18	3.2
MIP-Fair ($\alpha = 0.30$)	94	49.3	58.6	0.28	5.3
Pay-BRKGA ($\lambda_{\text{pay}} = 0.5$)	69	48.8	58.3	0.20	3.3

Table C.11: Metrics recorded in every run and not shown in the main tables: mean over the 24 instances (ten seeds averaged within each instance), except the maximum solve time, which is the maximum over all rounds and runs

Appendix C.5. Compliance and earnings by population

Table C.12 reports the compliance with the end-of-day guarantee by policy and regime, Table C.13 reports the end-of-day Gini coefficient by population discussed in Section 6.1, and Figure C.6 the distribution of end-of-day hourly earnings by policy discussed in Section 6.3.

Policy	**Manual**					**Auto-accept**				
	$\boldsymbol{\alpha = 0.30}$	**0.35**	**0.40**	**0.45**	**0.50**	$\boldsymbol{\alpha = 0.30}$	**0.35**	**0.40**	**0.45**	**0.50**
Greedy	**50**	**90**	**98**	**100**	**100**	**54**	**96**	**100**	**100**	**100**
ILP	0	0	0	18	52	0	0	4	46	58
Hierarchical	0	0	0	4	34	0	0	0	5	42
BRKGA	0	0	0	0	24	0	0	0	0	20
BRKGA-Fair ($\gamma = 40$)	0	0	0	8	46	0	0	0	5	46
MIP-Fair ($\alpha = 0.30$)	0	0	0	4	40	0	0	0	8	46
Pay-BRKGA ($\lambda_{\text{pay}} = 0.5$)	0	0	0	3	37	0	0	0	4	45
Second generation (five seeds)										
BRKGA-Fair ($\gamma = 40$, $k_d = 25$)	0	8	49	62	72	2	20	56	67	81
MIP-Fair ($\alpha = 0.30$, $k_d = 25$)	0	8	40	61	82	0	12	63	83	96

Table C.12: Compliance with the end-of-day guarantee: % of (instance, seed) pairs with $F_{\text{Gini}} \leq \alpha$, by regime

Appendix C.6. Ablations

Table C.14 reports the ablations of the supplement of X3, each paired with its base configuration on the same (instance, seed) pairs under the manual regime, together with the $k_d = 50$ rows of

Policy	Eligible couriers (connected ≥ 1 h) F_{Gini}, idle at zero	active only	Active couriers $F^{\text{act}}_{\text{Gini}}$	Active share of eligible	First quartile (COP/h, eligible)
Greedy	**0.301**	**0.279**	0.509	**0.97**	**481**
ILP	0.521	0.345	**0.414**	0.73	130
Hierarchical	0.527	0.387	0.530	0.77	146
BRKGA	0.565	0.376	0.488	0.70	86
BRKGA-Fair ($\gamma = 40$)	0.534	0.384	0.504	0.75	134
MIP-Fair ($\alpha = 0.30$)	0.544	0.366	0.460	0.72	105
Pay-BRKGA ($\lambda_{\text{pay}} = 0.5$)	0.546	0.364	0.457	0.71	97

Table C.13: The end-of-day Gini coefficient of hourly earnings by population, manual regime, mean over the 24 instances: the statistic of Eq. (7), its restriction to the eligible couriers that delivered, and the active-courier statistic $F^{\text{act}}_{\text{Gini}}$, which includes the couriers connected for less than one hour

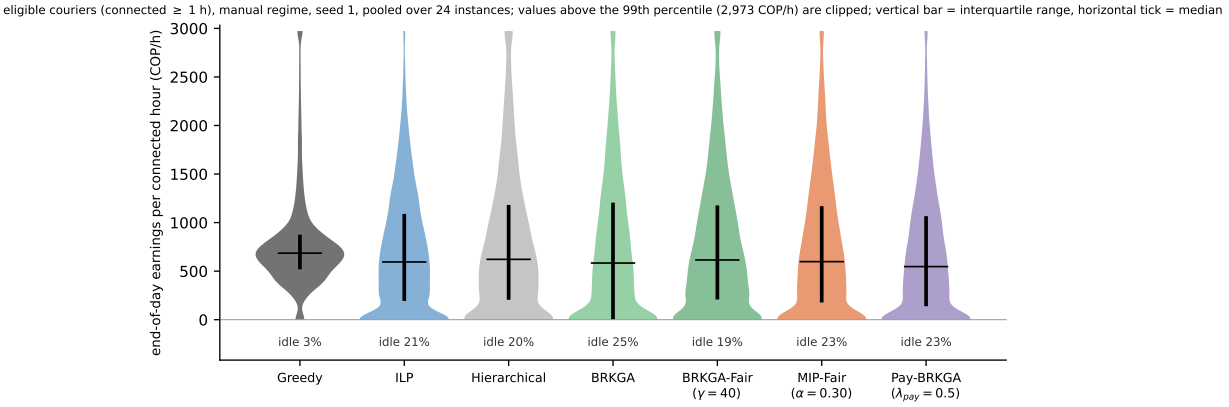


Figure C.6: Distribution of end-of-day hourly earnings over the eligible couriers by policy (manual regime, seed 1, 24 instances pooled; values above the 99th percentile clipped; bar = interquartile range, tick = median)

the second generation under both regimes. The table uses the five seeds of X3. The **MIP-Fair** rows with the free-only scope, the total-earnings basis and $k = 25$ have 115–116 pairs because their runs on the largest Bogotá day (instance 0) did not finish within the 240-minute timeout on four or five of the seeds, and the equity-first row has 27 pairs because the configuration was withdrawn from the grid after those runs (98 minutes per run for a policy that the pilot had already shown to be degenerate). The statistic to read is the paired difference with its 95% interval; the column "worse/better" counts the pairs in which the ablation ends more than 0.005 above or below its base in F_{Gini}.

Ablation (base)	n	**Coverage** mean	**Coverage** Δ base	**End-of-day Gini F_{Gini}** mean	**End-of-day Gini F_{Gini}** Δ base [95% CI]	**Worse / better**	**Active eligible**	**Gini $F^{\text{act}}_{\text{Gini}}$**	**CPU time** min (max)	**CPU** × base
Manual regime										
BRKGA decoder $\text{score}_{\text{finish}}$ (base: finish per order)	120	0.769	−0.001	0.558	−0.007 [-0.008, -0.005]	15/75	0.70	0.476	3.3 (17)	1.4
BRKGA 20×20 (base: 10×10)	120	0.769	−0.000	0.571	+0.007 [+0.005, +0.008]	76/8	0.69	0.501	6.2 (27)	2.7
BRKGA-Fair ($\gamma = 10$) total-earnings basis (base: hourly)	120	0.770	+0.000	0.534	−0.005 [-0.006, -0.003]	12/55	0.75	0.491	3.3 (17)	1.4
MIP-Fair Gini over free couriers only (base: all connected)	115	0.769	+0.006	0.498	−0.050 [-0.059, -0.041]	7/97	0.75	0.415	32.7 (183)	8.6
MIP-Fair total-earnings basis (base: hourly)	116	0.770	+0.007	0.515	−0.032 [-0.043, -0.022]	31/78	0.73	0.413	33.1 (240)	8.3
MIP-Fair equity-first priority (base: coverage first)	27	0.285	−0.469	0.153	−0.407 [-0.496, -0.317]	1/26	0.41	0.255	97.6 (227)	36.3
MIP-Fair prefilter $k = 25$ (base: $k = 8$)	115	0.770	+0.007	0.508	−0.039 [-0.047, -0.032]	9/98	0.74	0.404	30.6 (227)	8.0
MIP-Fair coverage slack $\varepsilon = 0.02$ (base: 0)	120	0.763	−0.001	0.543	−0.001 [-0.003, +0.000]	26/39	0.72	0.460	6.6 (35)	1.5
MIP-Fair coverage slack $\varepsilon = 0.05$ (base: 0)	120	0.763	−0.000	0.544	−0.000 [-0.002, +0.001]	27/24	0.72	0.463	6.6 (35)	1.5
BRKGA-Fair ($\gamma = 40$, exposure) $k_d = 50$ (base: $k_d = 25$)	120	0.768	−0.001	0.404	−0.033 [-0.036, -0.030]	0/118	0.88	0.374	6.4 (34)	1.3
MIP-Fair ($\alpha = 0.30$, exposure) $k_d = 50$ (base: $k_d = 25$)	120	0.763	−0.004	0.412	−0.023 [-0.028, -0.018]	17/89	0.87	0.447	51.1 (234)	2.2
Auto-accept regime										
BRKGA-Fair ($\gamma = 40$, exposure) $k_d = 50$ (base: $k_d = 25$)	120	0.786	+0.000	0.352	−0.064 [-0.068, -0.061]	0/120	0.91	0.323	5.5 (30)	1.3
MIP-Fair ($\alpha = 0.30$, exposure) $k_d = 50$ (base: $k_d = 25$)	120	0.786	+0.000	0.398	−0.002 [-0.007, +0.002]	44/67	0.88	0.427	22.4 (104)	2.0

Table C.14: Ablations of the supplement of X3, paired with the base configuration of each row on the same (instance, seed) pairs: number of pairs, coverage and F_{Gini} of the ablation with the paired difference with respect to the base, number of pairs worse / better than the base by more than 0.005 in F_{Gini}, active share of the eligible couriers, Gini coefficient over the active couriers, CPU time per run (mean and maximum, minutes) and its ratio to the base. Manual regime, five seeds of X3 (1,453 runs); the $k_d = 50$ rows also under auto-accept

BRKGA decoder and size. Scoring a chromosome by the finishing time instead of the finishing time per order lowers F_{Gini} by 0.007 and coverage by 0.001 at 1.4 times the CPU time. A population of 20 with 20 generations leaves coverage unchanged, raises F_{Gini} by 0.007 (worse in 76 of the 120 pairs) and costs almost three times the CPU time: the round objective is a coverage objective, and a longer search improves a quantity that the orders whose deadline can no longer be met already saturate while doing nothing for the end-of-day statistic. The 10×10 setting and the per-order score are retained.

*Scope and basis of the **MIP-Fair** constraint, and prefilter k.* Three ablations tighten the round constraint of **MIP-Fair**: the Gini over the free couriers only, the total-earnings basis and the prefilter with $k = 25$. The three lower F_{Gini} by 0.032–0.050 and raise coverage by 0.006–0.007, with 78–98 of the pairs better than the base. The expectation that the free-only scope would raise the end-of-day Gini because the free couriers are a biased subset is not borne out: F_{Gini} falls by 0.050 and the active-courier Gini from 0.365 to 0.330. None of the three reaches the second-generation variant with $k_d = 25$ ($F_{\text{Gini}} = 0.435$, Table C.9), and all three cost eight to nine times the CPU time of the base (31–33 minutes per day on average and up to four hours on a Bogotá day), which is consistent with rounds that run to the 10-second solver limit once the constraint binds.

A tighter round constraint is therefore not a substitute for the candidate set. The total-earnings basis in **BRKGA-Fair** moves F_{Gini} by 0.005: the weighted score reads the deficit only through the $K = 3$ candidates of the decoder, so the basis matters little, as the exposure basis of the second generation also showed.

Equity-first priority. The lexicographic order with equity before coverage, the first design of **MIP-Fair**, reproduces the collapse of the pilot on the 27 runs completed: coverage 0.285 against 0.754 for the base, a Gini of 0.153 obtained by not assigning (41% of the eligible couriers active, lateness ten minutes lower because few orders are served) and 36 times the CPU time. The paper's **MIP-Fair** keeps coverage first for this reason.

Coverage slack. The slack $\varepsilon \in \{0.02, 0.05\}$ leaves F_{Gini} and coverage within 0.001 of the base, with as many pairs worse as better; this is the first-generation counterpart of the ε result in Table C.9, and confirms that the lexicographic priority was not what bound the Gini block.

$k_d = 50$. The last two rows of each block answer the question left open in Section 6.1: whether the half of the gap to **Greedy** that remains at $k_d = 25$ is a matter of candidates or of the one-route-per-courier structure of the round. For **BRKGA-Fair** it is a matter of candidates: F_{Gini} falls from 0.437 to 0.404 under the manual regime and from 0.416 to 0.352 under auto-accept, no pair is worse than at $k_d = 25$, coverage is unchanged, 55% and 73% of the days meet $\alpha = 0.40$, and the CPU time grows by a third (6 minutes per day on average, 34 at most). The k_d curve of the heuristic, $0.534 \to 0.437 \to 0.404$ (manual) and $0.536 \to 0.416 \to 0.352$ (auto-accept) for $k_d \in \{0, 25, 50\}$, the $k_d = 0$ point being the activity-basis parent, is still descending. For **MIP-Fair** the curve flattens: $0.544 \to 0.465 \to 0.435 \to 0.412$ (manual) and $0.534 \to 0.426 \to 0.401 \to 0.398$ (auto-accept) for $k_d \in \{0, 8, 25, 50\}$, the last step within noise under auto-accept and with a coverage loss of 0.004 under the manual regime, while the CPU time grows to 51 minutes per day on average and almost four hours on the largest Bogotá day. For the exact round model the structure of the round and the solver limit bind before the candidate set does.

Appendix C.7. Gini coefficient under the sensitivity settings and emissions

Table C.15 is the companion of Table 3 for the end-of-day Gini coefficient over the eligible couriers. The statistic moves less than coverage under every factor: the regime lowers it by 2% for the first-generation policies and by 5–8% for the second-generation ones, whose deficit-aware prefilter reaches more idle couriers when every offer is accepted; the threshold moves it by at most 3.8%; α leaves it unchanged in the first generation.

Figure C.7 shows the regime loss discussed in Section 6.4, and Table C.16 the three criteria for the tolerated inequality α and the policy that satisfies each at the least coverage cost.

Table C.17 and Figure C.8 show the emissions diagnostic discussed in Section 6.4: distance and kg CO_2e per delivered order, the share of the distance driven by each vehicle type by policy, and

Policy\Setting	Relative difference of F_{Gini} over the base-case setting (in %)							
	Regime		Threshold θ (COP/km)			Tolerated inequality α		
	manual	auto	615	903	1,242	0.20	0.30	0.40
Greedy	0.0	−1.7	−0.7	0.0	+3.5	–	–	–
Hierarchical	0.0	−2.2	−1.2	0.0	+2.2	–	–	–
MIP-Fair ($\alpha = 0.30$)	0.0	−1.9	–	0.0	–	0.0	0.0	+0.1
BRKGA-Fair ($\gamma = 40$, $k_d = 25$)	0.0	−4.7	−2.3	0.0	+3.8	–	–	–
MIP-Fair ($\alpha = 0.30$, $k_d = 25$)	0.0	−7.8	−2.8	0.0	+3.2	–	–	–
Pay-BRKGA ($\lambda_{\text{pay}} = 0.5$)	0.0	−2.1	−1.4	0.0	+2.5	–	–	–

Table C.15: Sensitivity of the end-of-day Gini coefficient over the eligible couriers across operational settings: relative difference over the base case (manual regime, $\theta = 903$, $\alpha = 0.30$); sources and dashes as in Table 3

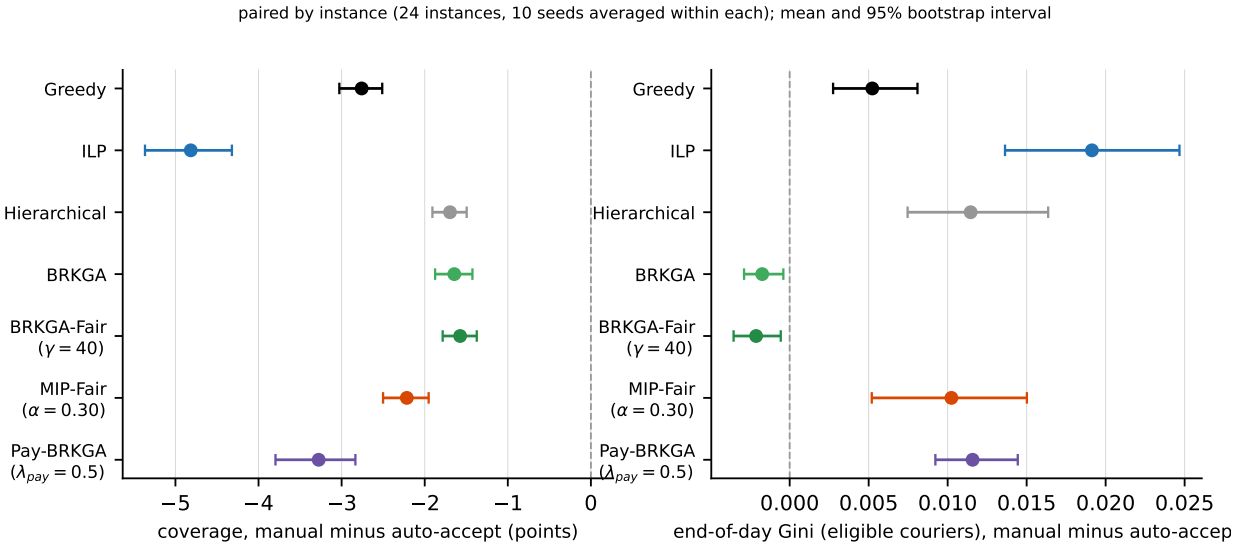


Figure C.7: Regime loss, manual minus auto-accept, paired by instance: coverage (left) and end-of-day Gini coefficient over the eligible couriers (right); mean and 95% bootstrap interval over the 24 instances, ten seeds averaged within each instance

Criterion	α	Policy satisfying it at least cost	Coverage cost vs. Greedy (pts)
A: historical (corporate Gini)	0.30 (X1: 0.301 manual, 0.296 auto-accept; 0.27–0.32 by city)	**Greedy** only; no order-centric policy meets it on any day	0 (status quo)
B: inequality band	0.30 (anchors 0.26 / 0.39); upper bound 0.40	at 0.40: **BRKGA-Fair** ($\gamma = 40$, $k_d = 25$) on 49% of the days, **MIP-Fair** ($k_d = 25$) on 40%	−1.1 / −0.9 (coverage gain)
C: Pareto knee	none: coverage flat within 0.3 points along α and along k_d	–	–

Table C.16: The tolerated inequality α under the three criteria and the recommended policy

the kg per order by policy and city, from the per-courier distances of the 240 manual-regime runs of each policy and the emission factors of Table C.5.

Policy	km per delivered order	kg CO_2e per delivered order	g CO_2e per km	Share of distance by vehicle (%) motorcycle	car	bicycle	walking
Greedy	2.94	0.280	95	73	7	17	3
ILP	2.33	0.216	93	72	6	17	5
Hierarchical	2.31	0.249	108	87	5	7	1
BRKGA	2.25	0.240	107	86	5	8	1
BRKGA-Fair ($\gamma = 40$)	2.27	0.237	105	84	5	9	1
MIP-Fair ($\alpha = 0.30$)	2.26	0.213	95	74	6	16	4
Pay-BRKGA ($\lambda_{\text{pay}} = 0.5$)	2.25	0.234	104	83	5	10	1

Table C.17: Emissions diagnostic by policy, manual regime: distance and emissions per delivered order and share of the distance by vehicle type, mean over the 24 instances (ten seeds averaged within each instance), from the per-courier distances and the emission factors of Table C.5

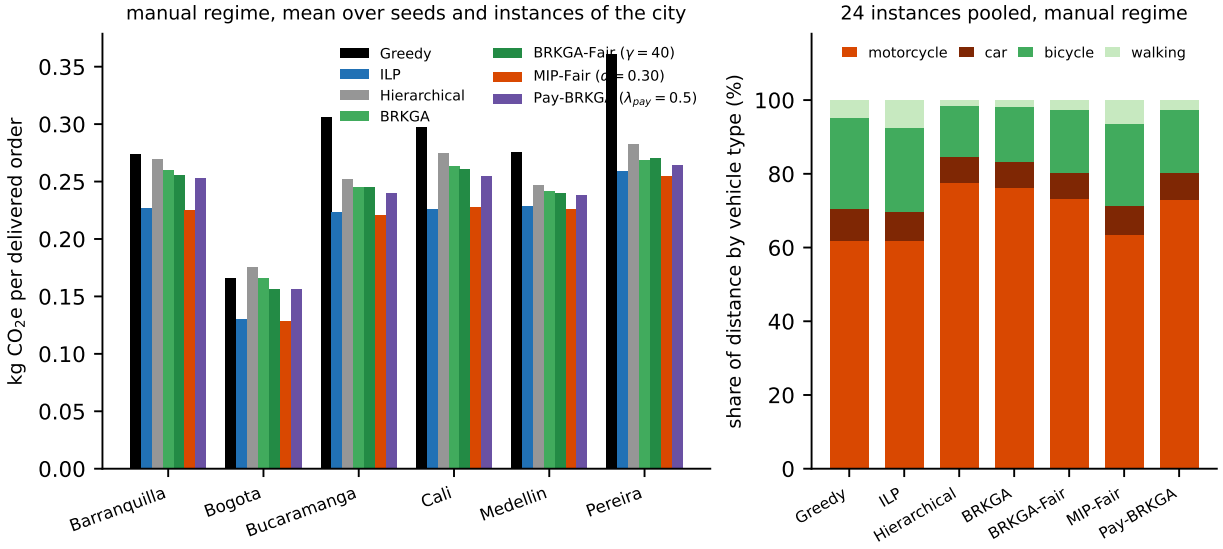


Figure C.8: Emissions diagnostic: kg CO_2e per delivered order by policy and city, manual regime, mean over the ten seeds and the instances of the city (left); share of the distance driven by each vehicle type, by policy, 24 instances pooled (right)